Highlights

Evaluating LLM-Based Credit Rating Systems: A Pre-Registered Specification-Curve Analysis

Anonymous

- In the evaluated systems, about 25% of matched elicitation pairs disagree on the investment-grade/high-yield classification.
- The elicitation axes jointly explain little rating variance (R^2 0.083–0.119). The fractional design does not identify per-axis shares. Pure seed noise is $\sim$8%.
- Instability persists on the temperature-0-honored subgrid (one provider, temperature 0).
- Unweighted cross-specification agreement stays modest (Fleiss $\kappa \leq 0.49$).
- A dispersion index predicts held-out instability (ROC-AUC 0.94) but is ordinally equivalent to counting dissenting elicitations, so it is reported descriptively.

Evaluating LLM-Based Credit Rating Systems: A Pre-Registered Specification-Curve Analysis

Anonymous[a]

[a]*Author affiliations withheld for double-blind review,*

ARTICLE INFO

Keywords:
large language models
credit ratings
specification-curve analysis
model risk
reproducibility
investment grade

ABSTRACT

Large language models (LLMs) are increasingly used to support credit assessments, but it is unclear whether elicitation choices merely reveal a stable credit view or shape the rating. Using a pre-registered specification-curve design that holds fundamentals fixed while varying how the assessment is elicited, we evaluate 45 credit-health profiles across 32 specifications and three seeds (4,320 credit-rating elicitations), drawn from a pre-registered 90-item battery and a balanced grid of model version, temperature, prompt paraphrase, output format, few-shot exemplars, and presentation over two providers, benchmarked to Altman Z″. Our estimand is stability, not accuracy. The elicitation axes jointly explain a small and model-dependent share of rating variance (R^2 between 0.083 and 0.119 across main-effects and two-way specifications). Because the fractional design aliases main effects with omitted interactions, we report no per-axis share. Pure run-to-run variation accounts for about 8%; instability persists on the temperature-0-honored subgrid (single provider, temperature 0, permutation $p = 0.0005$). Cross-specification agreement remains low at all resolutions. Unweighted Fleiss' κ reaches at most 0.49 at the investment-grade/high-yield split, where 25.2% of matched pairs disagree, and switching or pooling providers does not remove it. A small arm of 31 perturbed real issuers also exhibits specification variation, but we report it only as an illustrative descriptive limitation and draw no external-validity claim from it. Two pre-registered decontamination experiments (1,364 new elicitations, 1,361 substantive) are the reason. We do not detect evidence that absolute scale contributes additional instability under the paired within-issuer test ($p = 0.644$), but a closed-set probe identifies the issuer from its scale-invariant ratios alone at 19.4% against a 1/31 chance rate ($p = 5\times10^{-18}$), so the perturbation does not fully decontaminate. A post-hoc check does not detect attenuation on flagship models, and specification sensitivity is also observed in an open-weight reasoning model. We frame the elicitation specification as a model-risk parameter and report a specification-instability score that predicts held-out instability (ROC-AUC 0.94). Benchmarked against simple baselines, the score is *ordinally equivalent* to counting dissenting elicitations at the decision resolutions tested and carries no incremental discrimination over it, so we present it as a descriptive dispersion index and a governance trigger rather than as a validated control.

1. Introduction

Large language models (LLMs) are increasingly used as an analytic tool in finance, including for financial-statement interpretation, risk assessment, investment recommendations, and credit analysis (Hens and Nordlie, 2025). In such applications a user usually provides financial information and asks the model to make a judgment such as a credit rating, risk category, or trading recommendation. Implicit in such use is the presumption that the model has a sufficiently stable underlying assessment, and that reasonable changes to the phrasing of the question simply reveal that assessment. If this assumption is correct, then choices such as prompt wording, output format, sampling temperature, or model provider are mostly procedural. If it does not hold, then these choices are embedded in the measurement process itself.

This distinction is of particular importance for credit evaluation. Credit ratings are categorical decisions with economically important cut points. The investment-grade/high-yield split influences index eligibility, ratings-based investment mandates, treatment of collateral, covenant provisions, cost of financing, and regulatory capital. A model that reaches different assessments because the same financial information is presented under a different but defensible elicitation specification can lead to materially different downstream

decisions even when the underlying firm fundamentals are unchanged. In our experiments, around 25% of matched elicitation pairs place the same credit profile on opposite sides of the investment-grade/high-yield boundary. The key question we ask is therefore not whether an LLM can produce a plausible credit rating, but whether the rating is sufficiently stable when reasonable elicitation choices change.

Prompt sensitivity and run-to-run variability in LLMs have been well documented in computer science (Atil, Aykent, Chittams, Fu, Passonneau, Radcliffe, Rajagopal, Sloan, Tudrej, Ture, Wu, Xu and Baldwin, 2024; Kaplan, McCandlish, Henighan, Brown, Chess, Child, Gray, Radford, Wu and Amodei, 2020), and recent work has shown related variability in LLM-generated financial recommendations (Chon, Kim and Kim, 2025). But three questions remain unresolved for high-stakes credit applications. First, it is unclear how large this instability is in the units used by credit practitioners, rather than generic measures of textual variation. Second, the variation observed may be due to a small number of controllable design choices, in which case the instability could be mitigated through an appropriate choice of prompt or model configuration, or it may signal a broader noise floor that persists across the elicitation process. This design measures the overall magnitude of that sensitivity but does not identify how it is distributed across individual elicitation choices, so we do not attempt to settle that question. Third, evaluation using real firms poses a further challenge. General-purpose LLMs may recognize issuers from publicly available financial information, making it hard to distinguish independent credit assessment from memorized issuer knowledge.

We address these questions using a pre-registered specification-curve design based on the multiverse-analysis literature (Simonsohn, Simmons and Nelson, 2020; Steegen, Tuerlinckx, Gelman and Vanpaemel, 2016). Specification-curve analysis considers a single defensible analytical pipeline as a member of a larger family of reasonable specifications and tests whether the substantive result depends on the particular choice. Related approaches have been used to study researcher degrees of freedom and annotation variability (Camuffo, Gambardella, Kazemi, Malachowski and Pandey, 2026). We extend this logic to machine elicitation, where the elicitation specification is varied systematically while the financial information and the underlying task are held fixed. The distribution of ratings obtained is a direct measure of the dependence of the machine-generated credit opinion on the way it is asked.

This paper contributes an *elicitation-reliability evaluation framework* that measures how LLM-based credit ratings depend on the linguistic and procedural specification, together with a descriptive dispersion index and a simple review rule built on elicitation disagreement. In particular we (i) quantify the overall sensitivity of machine-generated credit ratings to the elicitation specification, reporting the joint fitted explanatory share of the elicitation axes, the identified provider-by-item component, pure seed variation, and residual variation, without attributing that share to individual axes; (ii) test the persistence of the instability under two natural remedies, the temperature-0-honored setting and provider selection, and across rating resolutions from the full letter scale to the investment-grade/high-yield split; (iii) translate the instability into the decision-relevant unit of investment-grade/high-yield boundary migration; (iv) run an illustrative descriptive real-issuer check on a perturbed arm that first demonstrates issuer-recognition risk and then mitigates exact-figure recognition through scaling and fingerprint screening, and (v) report a *specification-instability dispersion index* with pseudocode, a governance workflow, and a per-issuer cost model, and show that elicitation disagreement measured on one set of specifications predicts disagreement on a disjoint set (ROC-AUC 0.94). The index is descriptive. It is ordinally equivalent to counting dissenting elicitations at the decision resolutions tested, so we present disagreement as a simple governance and review trigger rather than as an instrument with incremental predictive value. The main experiment fixes 90 items and pre-registers a grid of elicitation choices, model version, temperature, prompt paraphrase, output format, few-shot exemplars, and answer presentation, evaluated across two providers over three repeated seeds (8,640 elicitations). The battery comprises two item families of 45 profiles each. This paper analyses the 45 credit-health profiles, giving the $45 \times 32 \times 3 = 4{,}320$ designed credit-rating elicitation cells, of which 4,316 returned substantive content and are analysed (Section 6.3), reported in the abstract and throughout Section 3. The 45 directional items are collected under the identical grid and released with the archive, but are not analysed here (Section 5.2). The credit-health profiles rest on a pre-specified Altman Z'' accounting-based benchmark (Altman, 1968), which provides a fixed, independently computable reference while the elicitation specification varies. Our estimand is therefore not rating accuracy but *stability across specifications*.

2. Related work

2.1. Prompt sensitivity and non-determinism in large language models

Large language models are built on transformer-based foundation models, and their output depends not only on the information provided to them but also on how that information is elicited (Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser and Polosukhin, 2017; Devlin, Chang, Lee and Toutanova, 2019; Bommasani et al., 2021). Prior work has demonstrated that semantically irrelevant or substantively minor changes to prompt construction, such as paraphrasing, answer format, ordering of options, and sampling configuration, can produce materially different outputs even when the underlying task is unchanged (Atil et al., 2024; Sclar, Choi, Tsvetkov and Suhr, 2024). Non-determinism can also be observed in settings intended to produce deterministic responses (Atil et al., 2024).

These results pose a significant measurement challenge for empirical applications of LLMs. If conclusions are sensitive to a single wording of a prompt, then an observed model response could be an artifact of the elicitation specification rather than a stable underlying assessment. Mizrahi, Kaplan, Malkin, Dror, Shahaf and Stanovsky (2024) demonstrate how single-prompt evaluation can produce fragile conclusions and argue for evaluation across multiple prompt formulations. In finance, Chon et al. (2025) likewise find large variation in stock recommendations produced by LLMs depending on the prompting and model choice.

2.2. Uncertainty quantification for LLM outputs

A parallel literature seeks to quantify how confident a model's output is. Self-consistency samples a model repeatedly and aggregates by majority vote, improving reasoning accuracy and exposing answer dispersion (Wang, Wei, Schuurmans, Le, Chi, Narang, Chowdhery and Zhou, 2023). Semantic-entropy methods cluster semantically equivalent generations and measure entropy over meanings to detect unreliable answers (Kuhn, Gal and Farquhar, 2023). Others elicit confidence directly, teaching models to verbalize calibrated uncertainty (Lin, Hilton and Evans, 2022) or probing whether models "know what they know" (Kadavath, Conerly, Askell, Henighan et al., 2022). These methods largely target open-ended generation and token-level or free-text confidence. Our specification-instability score is a complementary uncertainty measure aimed at a *categorical decision*. Rather than sampling one prompt many times or reading token probabilities, it measures dispersion of the decision class across a fixed set of defensible elicitation specifications, requires no ground truth or model internals, and reports directly in the units of the downstream decision (for example, the IG/HY boundary).

2.3. Multiverse analysis and specification curves

Empirical results often depend on a set of reasonable analytical choices rather than a single, uniquely correct specification. The problem has been described as the "garden of forking paths," in which several defensible decisions can lead to different reported results even in the absence of intentional specification search (Gelman and Loken, 2014; Simmons, Nelson and Simonsohn, 2011). Multiverse analysis and specification-curve analysis tackle this problem by assessing the effect across a specified set of reasonable analytical specifications rather than a single chosen pipeline (Steegen et al., 2016; Simonsohn et al., 2020).

Specification-curve analysis is particularly useful when the emphasis is on how sensitive a result is to the choices made by the researcher. The approach examines the distribution of outcomes over the entire set of defensible specifications, rather than asking whether a given specification leads to a desired outcome. Similarly, Camuffo et al. (2026) introduce a variance-aware approach for LLM-assisted management annotation, underlining the importance of accounting for the variability introduced by machine-generated annotations.

We extend this framework to LLM elicitation. We treat each combination of the model and the prompting choices as one specification, while holding the financial information provided to the model fixed. We treat the LLM as the estimator under study, and the specification curve that arises measures the degree to which the inferred credit assessment varies solely due to reasonable choices in how that assessment is solicited.

2.4. Large language models for finance, credit, and supervision

LLMs are increasingly employed for financial tasks that were previously the preserve of specialist judgment, traditional statistical models, or structured quantitative systems. General-purpose (Brown et al., 2020; Ouyang et al., 2022; OpenAI, 2023) and finance-specific (Wu, Irsoy, Lu, Dabravolski, Dredze,

Gehrmann, Kambadur, Rosenberg and Mann, 2023; Xie, Han, Zhang, Lai, Peng, Lopez-Lira and Huang, 2023) language models have been evaluated for return prediction (Lopez-Lira and Tang, 2023), analysis of central-bank communication (Hansen and Kazinnik, 2024), risk profiling (Hens and Nordlie, 2025), banking supervision (García-Llorente and Olmeda, 2026), and credit-rating prediction (Drinkall, Pierrehumbert and Zohren, 2025).

Much of this literature evaluates *capability*, meaning whether an LLM can extract useful financial information, produce accurate forecasts, classify firms, or approximate expert judgments. For instance, recent work on credit-rating forecasting compares generative LLMs with established predictive approaches and finds that conventional methods can retain an accuracy advantage (Drinkall et al., 2025). These studies concern whether the model performs the task well.

Our question comes before that comparison. An LLM-generated credit assessment should be sufficiently reproducible under reasonable elicitation choices to be interpretable as a meaningful measurement. A model can appear accurate under one prompt and output a materially different rating under another equally defensible prompt. We therefore emphasize *stability versus capability*, meaning whether the credit opinion itself is unchanged when the information being evaluated is unchanged but the elicitation specification differs.

2.5. Credit ratings, economic consequences, and model risk

Quantitative credit-risk assessment has a long history. Altman's bankruptcy-prediction framework (Altman, 1968) and later probabilistic approaches such as Ohlson's model (Ohlson, 1980) showed that accounting information contains systematic signal about corporate financial distress. Credit ratings encode economically relevant differences across firms, and disagreement among human rating agencies has been studied as a meaningful feature of the credit-rating process itself (Cantor and Packer, 1997; Kliger and Sarig, 2000; Tang, 2009).

Rating stability matters because shifts in rating categories can influence financing terms and institutional choices. Rating differences are associated with borrowing costs and market pricing (Kliger and Sarig, 2000; Tang, 2009; Cornaggia, Cornaggia and Israelsen, 2018), and migration across the investment-grade/high-yield boundary can have particularly important implications for investment mandates, portfolio composition, and capital treatment. On the regulatory side, the Basel standardized approach maps external credit assessments into risk weights for capital calculations (Basel Committee on Banking Supervision, 2019), and supervisory exercises have documented considerable dispersion in risk-weighted assets even for common portfolios (Basel Committee on Banking Supervision, 2013).

These concerns connect LLM elicitation to the broader literature on model risk. Under supervisory guidance, models must be validated, their limitations understood, and the uncertainty in their outputs appropriately characterized (Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency, 2011). If the elicitation specification for an LLM-generated credit rating changes the rating materially, then the specification itself becomes part of the model-risk environment. Our work ties these literatures together by directly measuring that instability and quantifying it both through statistical disagreement and through investment-grade/high-yield rating migration, the decision-relevant unit of the categorical credit limit.

3. Results

The pre-registered design assesses the stability of ratings across two providers (OpenAI and Google), each evaluated at a current and a prior model snapshot (four snapshots in total), giving balanced coverage across providers, versions, and the elicitation axes. Exact snapshot identifiers are listed in the Methods (Table 12).

3.1. Sources of rating variation

We first consider the source of variation in the machine-generated credit rating. An analysis-of-variance decomposition of the full letter-rating rank (ordinary least squares with Type-II sums of squares) attributes variance to the firm profile, individual elicitation-design axes, the provider-by-item interaction, and a residual term (Table 1).

As expected, the firm profile explains most of the observed variance, about 65% (95% CI 54–72%), indicating that the models respond strongly to differences in the underlying financial information. But the

remaining variation is important because the financial information is constant within each item. Jointly, the elicitation axes explain little of it. The design matrix supports an R^2 of 0.083 under main effects only and 0.119 with all two-way interactions added (Table 2).

We do not report individual per-axis shares. On the analysed 32-specification two-provider design, main effects are mutually near-orthogonal (maximum off-diagonal $|r| = 0.125$) but are aliased with two-factor interactions of other factors up to $|r| = 0.504$, so each main-effect sum of squares is identified only up to a contribution from omitted interactions whose magnitude the design cannot bound without an external assumption (Section 6.2 gives the alias algebra and the norms). One component *is* identified and is reported as such. Because items are fully crossed with specifications, the provider-by-item interaction is effectively orthogonal to the specification-axis contrasts up to the numerically negligible imbalance quantified in Section 6.2, and it contributes ∼1.4% of variance. The OLS residual is ∼25%. Because a raw residual can conflate genuine run-to-run randomness with structure the model omits, we separate the two: *pure within-cell seed variation*, the variance across the three seeds of each identical item×specification cell, accounts for only 8.1% of total variance (95% CI 6.4–10.8%), leaving the remaining ∼17% as variation not explained by the main-effects model. That part of the residual is not identified as to source under this resolution-limited D-optimal design (see Limitations). It may combine real unmodeled interactions with main effects aliased into two-factor interactions, so we interpret it descriptively instead of attributing it to structured sensitivity. Run-to-run sampling noise is a real but minor part of the residual.

We report the variance the elicitation axes explain under three alternatives (Table 2): a main-effects-only model, a model adding all two-way interactions, and an issuer-out-of-sample regularized version of the latter. The apparent explained variance is dependent on the modeling choice, but is low for all three (R^2 between 0.08 and 0.12), and the in-sample gain from interactions does not fully generalize out of sample, confirming that individual-axis attribution is not identified and that the decomposition should be read in a descriptive way.

Under the fitted models, the elicitation axes jointly explain a small and model-dependent share of rating variation ($R^2 = 0.083$–0.119). The fractional design does not identify how that share should be attributed to individual axes, so we make no claim about whether the sensitivity is concentrated in one choice or distributed across several.

The same pattern appears in the specification curve (Fig. 1). The three seeds are pooled, and each point is a pre-registered specification, sorted by how consistently it recovers the constructed Altman Z″ design labels. Because the profiles were constructed to satisfy those strata, we interpret this panel as a diagnostic of design-label recovery, not external credit-rating accuracy. Even when the basic firm profiles are the same, label recovery varies quite a bit across defensible specs. Specs split approximately 56/44 around the pre-registered majority threshold. Thus, the specification curve shows that conclusions drawn from a single elicitation configuration do not necessarily generalize to other plausible configurations.

3.2. Specification fragility on the temperature-0-honored subgrid

One explanation of this residual variation is that a provider may not fully honor a requested temperature of zero. If so, the observed instability might largely reflect stochastic sampling rather than sensitivity to the elicitation specification per se.

We therefore perform the pre-registered determinism test and restrict the analysis to the provider-setting combination for which the returned metadata confirm that temperature was honored, namely Google at temperature 0. This temperature-0-honored subgrid sharply reduces repeated-call sampling variation but retains substantial specification dependence. In it, 62.5% of specifications are on opposite side of the majority label-recovery threshold, and consistency with the constructed credit-band design labels ranges from 0.36 to 0.69 across temperature-0 prompt configurations.

A one-sided permutation test rejects the null of specification invariance of the *rating* itself. The test permutes specification-condition assignments across profiles (a null of rating invariance, rather than permuting accuracy labels), with the statistic the between-specification spread of the item-centred letter grade. On the temperature-0-honored Google subgrid, the observed spread is 1.59 notches, versus a permutation-null mean of 0.18 notches and standard deviation 0.050 notches. Across 2,000 permutations none achieved the value observed, giving $p = 0.0005$ (specification $\eta^2 = 0.46$ on the item-centred rating, a specification-level aggregate not comparable to the individual-rating partition of Table 1). Holding

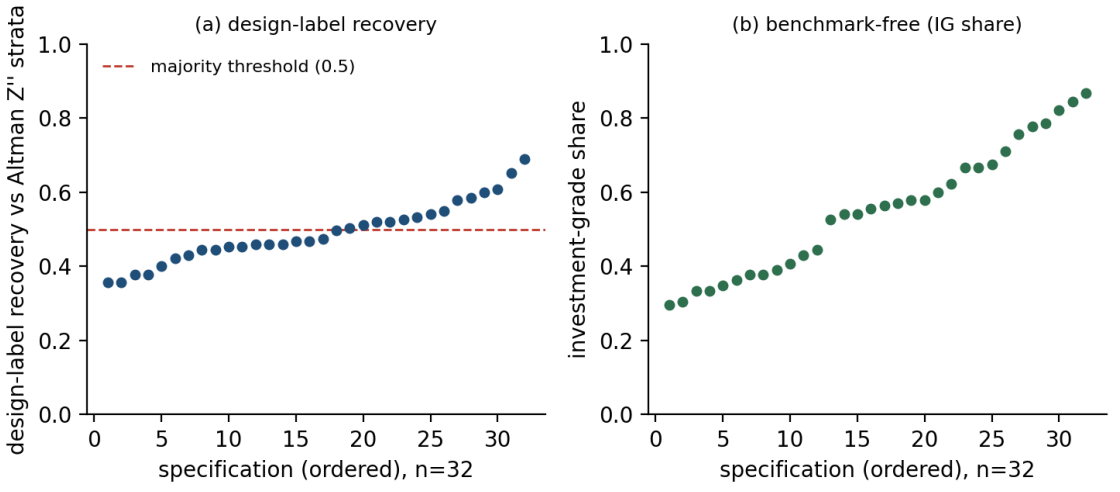


Figure 1: Specification curves of the machine credit rating. Each point is one pre-registered specification (three seeds pooled), specifications sorted independently in each panel. **(a)** Design-label recovery: agreement with the constructed Altman Z″ strata (the profiles were built to satisfy these strata, so this is label recovery, not external accuracy) the specs split about 56/44 around the majority line. **(b)** Benchmark-free: the share of a specification's credit elicitations rated investment grade, which needs no ground truth and ranges widely across specifications, direct evidence that the verdict, not merely its accuracy, moves with the elicitation. This is the multiverse the method is named for.

Table 1
Variance shares of the machine credit letter-rank (OLS decomposition with Type-II ANOVA, $n = 45$ items). The residual is split into pure within-cell seed variance and other unexplained variance. Confidence intervals (profile-clustered bootstrap, 95%) are reported only for the item and pure-seed components. Individual elicitation-axis shares are not reported because the fractional design does not identify them; the joint explanatory share of the elicitation axes is reported in Table 2 (Section 6.2).

Component	Variance share
Item (firm profile)	64.9% (53.9–72.3%)
Residual (total)	25.2%
— pure within-cell seed variation	8.1% (6.4–10.8%)
— other unexplained (unmodeled interactions)	∼17%
Elicitation axes, jointly	see Table 2
Provider × item interaction	1.4%

The two indented rows decompose the residual and are not additional components. Individual elicitation-axis shares are not reported because the fractional design does not identify them; the joint explanatory share of the elicitation axes is reported in Table 2. The rows shown do not form a complete partition of the total variance. Design diagnostics on the analysed 32-specification two-provider design: the maximum off-diagonal correlation between main-effect columns of distinct axes is 0.125, and the maximum absolute correlation between a main-effect column and a two-factor-interaction column of other axes is 0.504 (Section 6.2). As a quality-control check we re-audited all design-dependent diagnostics and variance-decomposition inputs on this design, and no other design-dependent diagnostic or variance-decomposition result changed.

temperature at 0 thus removes much of the run-to-run jitter but does not eliminate sensitivity to prompt and presentation choices. Indeed, the specification-curve flip share within the temperature-0-honored subgrid is higher than the pooled estimate. The instability therefore cannot be ascribed solely to stochastic sampling, consistent with the variance decomposition, in which pure seed variation is a minority of the residual.

3.3. Agreement within and between providers

Provider choice might be thought immaterial. Direct agreement comparisons reveal a more complex pattern.

Table 2
Variance in the machine letter rating explained by the elicitation axes under different model specifications (credit-health task, issuer identity removed to isolate the axes). The estimate is changing with the model and remains small throughout. This illustrates that per-axis attribution is not identified under the design.

Model specification	R^2
Main effects only	0.083 (in-sample)
Main effects + all two-way interactions	0.119 (in-sample)
Regularized two-way (ridge), issuer out-of-sample	0.101 (5-fold, issuer group)

Within-provider agreement is consistently higher than cross-provider agreement. Within a single provider, 82.5% of identical firm profiles agree at the three-way credit-band level, but only 64.4% agree across the two providers. At the full letter-rating level, agreement across providers falls to 11.1% from 30.4% within providers.

These findings are consistent with the variance decomposition. The two providers respond differently to specific firm profiles, creating a provider-by-item interaction. The interaction does not appear to reflect a constant directional shift in ratings. A simple global adjustment does not allow one provider to be characterized as systematically more conservative or more generous than the other. The disagreement is issuer-specific.

This distinction is important operationally. Changing vendors does not provide a predictable correction to the rating of a specific firm, and aggregating outputs across vendors does not remove the underlying specification dependence. Vendor diversification therefore does not resolve the conceptual measurement problem or the need to characterize uncertainty across elicitation specifications.

3.4. Stability across rating resolutions

Credit ratings are inherently hierarchical. A model may not reproduce an exact letter grade yet still express a stable broader assessment. We therefore assess cross-specification agreement as we move from the full 21-grade letter-rating scale to notch categories, three broad credit bands, and the binary investment-grade/high-yield distinction (the deterministic mappings are given in Table 11).

Agreement improves as the scale is coarsened but does not become fully stable. Raw cross-specification agreement is approximately 17% at the letter level, 36% at the notch level, 68% for the three-way credit band, and 75% for the investment-grade/high-yield split. Equivalently, matched elicitation pairs disagree in about 83%, 64%, 32%, and 25% of comparisons, respectively (Fig. 2).

The same conclusion follows from chance-corrected agreement. Fleiss' κ is 0.10 at the full letter scale, 0.21 at the notch level, 0.43 for the three-way band, and reaches a maximum of only 0.49 for the investment-grade/high-yield split (Fig. 3). No evaluated rating resolution meets the conventional threshold for substantial agreement ($\kappa = 0.60$) under unweighted scoring.

Because unweighted κ on 21 ordinal grades penalizes an AAA-versus-AA+ disagreement as heavily as AAA-versus-C, we also report ordinal-weighted agreement on the letter-rank scale. Krippendorff's α is 0.44 under linear weighting and 0.65 under quadratic weighting. Fine-grained disagreement is therefore concentrated among adjacent grades. This weighting, however, cannot apply to the binary investment-grade/high-yield decision, where the only two categories are maximally distant and 25.2% of matched pairs still cross the boundary. The decision-relevant instability is unaffected by how ordinal proximity is credited.

The instability is not caused by the models collapsing to a narrow set of central ratings. All 21 letter grades are used across the experiment (pooled scale-usage entropy 3.89 of a maximum 4.39 bits). This spread is not merely across firms. The mean *per-firm* rating entropy is 2.85 bits, so a single firm's rating is itself distributed over roughly seven effective grades as the specification varies. The models therefore use the available rating scale extensively but do not reproduce the same assessment consistently across specifications.

The most economically consequential result appears at the coarsest resolution. Even when all finer distinctions are dropped and only the investment-grade/high-yield classification is retained, 25.2% of matched elicitation pairs place the same firm profile on opposite sides of the boundary. Coarsening reduces specification sensitivity but does not eliminate it.

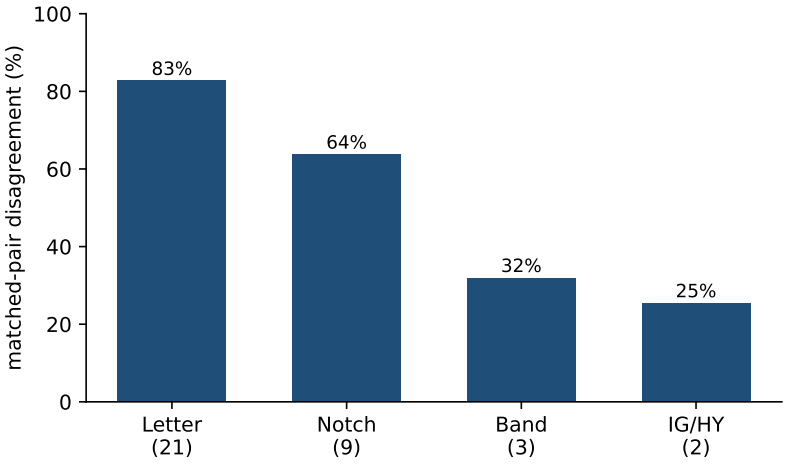


Figure 2: Share of matched elicitation pairs whose credit verdicts differ, by rating granularity, where the two elicitations of a firm may differ on any grid axis (provider, model version, temperature, paraphrase, format, few-shot, presentation) and seed. Even at the coarse investment-grade/high-yield split, the most consequential line in credit, about 25% of matched pairs disagree. Matched pairs are drawn independently *with replacement* from a profile's own elicitations, $n_i = 96$ for 41 profiles and 95 for the four that are short one seed. The two draws may fall in the same specification×seed cell, in which case the pair agrees, and may differ in specification, in seed, or in both, while the firm profile is always held fixed (Section 6.3).

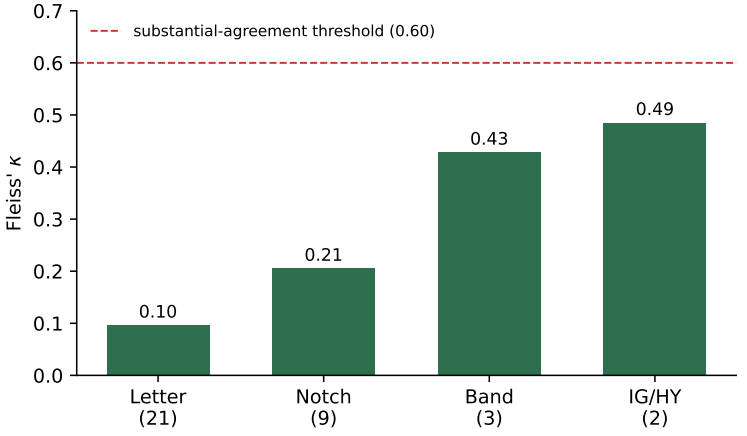


Figure 3: Chance-corrected cross-specification agreement (unweighted Fleiss κ) by granularity. Agreement peaks at 0.49 on the IG/HY split and no resolution reaches the $\kappa = 0.6$ substantial threshold under unweighted scoring. Ordinal weighting raises letter-scale agreement (Krippendorff's $\alpha = 0.65$ quadratic), but the binary IG/HY flip is unaffected by weighting.

This boundary instability is not uniform across the credit spectrum. The per-comparison IG/HY flip is 13.0% among benchmark-SAFE profiles, rises to 35.5% among benchmark-WATCH profiles adjacent to the boundary, and is 27.1% among benchmark-DISTRESS profiles. Because the battery is balanced across the three strata by construction (15 profiles each), the pooled 25.2% is the average over strata rather than an artifact of oversampling near-boundary firms. The flip is elevated in the benchmark-WATCH stratum, but when we treat distance from the boundary continuously it does not decline monotonically with that distance (Spearman between per-item flip and $|Z'' - 2.60|$ is -0.05), so the instability is broadly distributed across the credit spectrum rather than confined to a narrow near-boundary band.

Table 3
Fragility in decision and rating units. Parenthetical values are 95% confidence intervals (profile-clustered percentile bootstrapping)

Quantity	Result
Specifications flipping majority credit *band* verdict (pooled / determ.)	∼56% / 62.5%
Design-label recovery range, temperature-0-honored subgrid	0.36–0.69
Permutation test, temperature-0-honored subgrid (rating-invariance null)	$p = 0.0005$ (2,000 perms)
Within- vs cross-provider agreement (band / letter)	82.5/64.4 · 30.4/11.1
Max chance-corrected agreement (Fleiss κ, IG/HY)	0.49 (no scale ≥ 0.6)
Per-comparison IG/HY boundary flip	25.2% (95% CI 19.7–31.2%)

3.5. Migration from investment-grade to high-yield

Statistical disagreement is economically important when alternative specifications produce decisions on opposite sides of material credit boundaries. The observed rating instability is understood as the decision-relevant unit of investment-grade/high-yield migration.

For each credit-health profile we draw two elicitations from its observed specification distribution, allowing the pair to differ on any elicitation axis and on seed. We then compute, for each firm, the probability that the two resulting ratings fall on opposite sides of the investment-grade/high-yield boundary, and average across profiles. This per-comparison flip probability is one minus the Gini–Simpson concentration of the firm's binary IG/HY outcomes.

This yields a total investment-grade/high-yield elicitation disagreement rate of 25.2% (95% CI 19.7–31.2%), in which the two draws may differ in specification, in seed, or in both. For context, we report two complementary conditional measures: 24.3% after collapsing repeated seeds within specification, and 6.9% when varying seed within an identical profile×specification cell. These quantities are not additive components of the 25.2% total. Forming the per-specification modal outcome requires a tie rule, and we treat a cell with no modal winner as dissenting; the rule never binds here, because all four two-observation cells are unanimous and no three-observation cell lacks a modal winner. The headline figure is therefore total elicitation instability and is not attributed to specification choices alone: about a quarter of matched ratings for the same underlying credit profile disagree on which side of the investment-grade boundary the profile falls (Table 3). Because the boundary governs index eligibility, mandate constraints, collateral treatment, and cost of financing, migration of this magnitude driven by the elicitation process, with the fundamentals held fixed, is economically consequential.

3.6. Illustrative check on perturbed and fingerprint-screened real issuers

The constructed battery provides the controlled context in which to identify elicitation instability, but a key external-validity concern is whether similar behavior appears for real firms. Direct evaluation on real issuers raises a further methodological problem. An LLM may recognize a company from publicly available financial figures even when the company name is removed.

We test this possibility, and then use the real issuers as an illustrative descriptive check only, not as an external validation. Section 3.7 reports two pre-registered decontamination experiments whose combined result requires that restriction. When presented with anonymized but otherwise unchanged financial profiles, Gemini-Flash correctly identifies about 22% of the selected mid-cap issuers from their financial numbers alone. This shows that anonymizing the issuer name does not guarantee that an LLM makes an independent credit assessment rather than relying on memorized issuer information.

The pre-registered fingerprinting gate lowered the issuer-identification rate from 22% to 2.9%, below the pre-specified 5% threshold. The retained perturbed firms are then rated on a frozen 12-specification grid at three seeds, and the constructed battery is re-rated under exactly the same specification grid to provide a matched comparison.

The real-firm WATCH-stratum flip share is 0.226 (95% CI 0.147–0.308), relative to 0.318 (95% CI 0.223–0.406) for the matched constructed battery. The estimated difference is −0.091 (95% CI −0.213 to 0.032, Table 4), not statistically distinguishable from zero. We treat this arm as an illustrative descriptive real-issuer check, and specification variation is also observed in the perturbed and fingerprint-screened real

Table 4
Real-firm arm: 31 perturbed and fingerprint-screened business-to-business issuers, agency-rating benchmark. Parenthetical values are 95% confidence intervals (issuer-clustered percentile bootstrap).

Quantity	Result
Fingerprint identification, naive (exact figures)	22%
Fingerprint identification, after perturbation and screening	2.9%
WATCH-stratum flip-share, real arm	0.226 (0.147–0.308)
WATCH-stratum flip-share, battery (same 12 specs)	0.318 (0.223–0.406)
Difference (real − battery), 95% CI	−0.091 (−0.213, 0.032)
external-benchmark spec. curve, per-spec accuracy range	42.2–61.1% (spread perm. $p = 0.021$)
Interpretation	illustrative descriptive real-issuer check
Machine band vs. disclosed agency rating (descriptive)	53.7%
marginal-matching chance baseline	48.9%
excess over that baseline, 95% CI	+4.9pp (−7.9, +17.1); $\kappa = 0.095$

issuers under the same elicitation conditions, but the small sample (16 real, 15 battery WATCH observations) cannot establish that its magnitude equals the constructed battery's.

We also discuss the relevant concern that the perturbation-and-screening procedure or the selection of the external benchmark may be driving the result. First, the per-issuer scaling is uniform, meaning all financial ratios and the Altman Z″ are invariant by construction (Z″ transformed-vs-original correlation 1.000000, 100% of issuers retain their original stratum), and the realized scale factors, which span the full pre-registered $[0.6, 1.7]$ range across the 31 issuers, are not analysed against per-issuer instability here, and because each issuer received a single scale draw, scale cannot be separated from issuer identity in this comparison; Section 3.7 instead tests scale within issuer. Second, using the disclosed agency ratings as an external benchmark not used to construct the profiles, per-specification credit-band accuracy on the real issuers (the SAFE and WATCH strata, i.e. 30 of the 31 issuers, the single DISTRESS issuer is a singleton excluded from this three-band accuracy comparison) ranges from 42.2% to 61.1% across the 12 specifications (mean 53.7%); an issuer-preserving permutation test (specification labels permuted within issuer) finds this between-specification spread beyond chance. Per-specification agreement with the external agency benchmark therefore varies across the evaluated specifications ($p = 0.021$). This is a within-issuer specification contrast and is not interpreted as evidence of rating accuracy. Machine-generated credit bands correspond to disclosed agency ratings 53.7% of the time. The correct reference for that figure is not the uniform three-class value of 33.3%: both margins are concentrated in two bands. The machine assigns SAFE 54.7% and WATCH 43.7% of the time, the agency benchmark 46.7% and 53.3%, so the agreement expected from independent draws with these margins is $\sum_c p_{\text{machine}}(c)\, p_{\text{agency}}(c) = 48.9\%$. Measured against that baseline the excess is +4.9 percentage points with an issuer-clustered 95% interval of (−7.9, +17.1), and Cohen's κ is 0.095. **Machine bands are therefore not distinguishable from chance agreement with the agency benchmark once the observed margins are accounted for**, and we draw no accuracy claim from this comparison. What the real-firm arm is used for is the *specification spread* reported above, which is a within-issuer contrast and does not depend on the level of agreement.

3.7. Decontamination of the real-issuer arm: two pre-registered experiments

Screening the real-issuer arm for *exact-figure* recall and applying one per-issuer scale draw leaves two distinct objections unaddressed. A uniform scaling preserves every financial ratio by construction, so the *ratio pattern* may itself identify the issuer, and with one draw per issuer the scale factor is confounded with issuer identity. We therefore pre-registered and ran two experiments, hashing the specification before any call (`SPEC_r3_realarm_experiments.md`, deposited with the archive). In total 1,364 new elicitations across both providers, of which 1,361 returned substantive content (three empty completions, all on one provider, are dropped listwise as in the main corpus). Table 5 reports both at their value, and they point in opposite directions.

Experiment B returns a null. Re-eliciting every issuer at scale factors 0.6, 1.0 and 1.7 on the same frozen 12-specification grid makes scale vary *within* issuer, so each issuer is its own control. The three conditions

Table 5
Decontamination experiments for the real-issuer arm, pre-registered and hashed before collection. 1,364 elicitations attempted and frozen, 1,361 substantive, two providers, 31 issuers. The "calls" column reports substantive completions. Intervals are issuer-clustered bootstrap (2,000 draws). The bootstrap unit is the issuer. Experiment A's closed-set probe supplies the 31 true names as candidates and is therefore an upper bound on identifiability rather than a realistic leakage estimate.

	calls / pairs	rate	95% CI
Experiment A: can the ratio pattern identify the issuer?			
Open-ended ("name the company")	60	0.000	(0.000, 0.000)
Closed-set, 31 candidates	185	0.194	(0.124, 0.274)
chance (1/31)	—	0.032	—
exact binomial vs chance	—	$p = 5.2e-18$	—
issuers identified at least once	—	19/31	—
issuers identified in every ordering	—	0/31	—
Exact-figure fingerprint gate (Section 3.6)	—	0.029	—
Experiment B: does absolute scale move the rating within issuer?			
IG/HY class differs, scale 0.6 vs 1.7	372	0.159	(0.113, 0.207)
IG/HY class differs, scale 0.6 vs 1.0	372	0.156	(0.118, 0.196)
IG/HY class differs, scale 1.0 vs 1.7	372	0.142	(0.100, 0.183)
permutation null mean	372	0.152	—
within-issuer permutation test	—	$p = 0.644$	—

are multiples of the arm's *already perturbed* figures, not of the original filings: the frozen battery applies a per-issuer LogUniform(0.6, 1.7) draw with ratios preserved, the 1.0 condition reproduces those perturbed figures unchanged, and 0.6 and 1.7 rescale them further. The real unscaled figures exist only in the sealed crosswalk and never enter a prompt. The 1.0 condition therefore carries no claim about contamination, and the decontamination-relevant contrast is 0.6 versus 1.7, a within-issuer factor of 2.83 applied to a common baseline. The IG/HY class differs between the extreme scales for 15.9% of (issuer, specification) cells, but the within-issuer permutation null, in which scale labels are exchangeable, sits at 15.2%. The permutation test does not reject ($p = 0.644$, 10,000 permutations). We therefore do not detect evidence that absolute scale contributes additional instability beyond the re-elicitation variability this null represents. Non-rejection is not proof that no scale effect exists. The comparison is a properly paired within-issuer test, so scale is not confounded with issuer identity.

Experiment A detects residual identification and therefore limits how the real-issuer arm can be interpreted. Given only scale-invariant ratios and asked to name the company, the models identify *none* of the 31 issuers, with zero hits in the 60 substantive completions of 62 attempted. Given the same ratios together with the 31 true issuer names in randomised order, they select correctly in 19.4% of calls (95% CI 12.4–27.4%) against a chance rate of $1/31 = 3.2\%$, an exact binomial $p = 5\times10^{-18}$. 19 of 31 issuers are identified at least once. The signal is real but unstable. No issuer is identified under all three candidate orderings, and the open-ended probe recovers nothing, so this is recognition assisted by a candidate list rather than reliable recall. A closed-set probe that supplies the answers is an *upper bound* on identifiability, not an estimate of realistic leakage. It is nonetheless above chance, and our pre-registered decision rule required the closed-set rate to be indistinguishable from 1/31 with a bootstrap upper bound below the 5% screening threshold. It is not. Because the prespecified decontamination criterion is not met, we treat the real-issuer arm as an illustrative descriptive limitation and draw no external-validity claim from it. Scale-invariant ratios are evidently identifying enough that uniform scaling cannot be treated as decontamination on its own, which we think is useful beyond this paper, since ratio-preserving perturbation is the obvious design for anyone evaluating a model on real firms.

Table 6
Model-tier robustness (post-hoc, not pre-registered): per-comparison flip share on the same frozen 12-specification grid (45 credit-health items, three seeds). Parenthetical values are 95% profile-clustered bootstrap confidence intervals. The difference is paired by item.

Metric	Nano/flash	Flagship	Difference (95% CI)
IG/HY flip (pooled)	24.7% (19.1–30.6)	23.6% (19.4–27.8)	−1.1 (−7.6, +6.0)
IG/HY flip, OpenAI	17.3%	26.7%	
IG/HY flip, Google	20.5%	16.7%	
Three-way band flip	30.5%	25.2%	−5.3 (−11.9, +1.3)
Design-label recovery vs Altman	44.3%	36.7%	
IG rate (leniency)	58.6%	79.9%	

Nano/flash: `gpt-5.4-nano` + `gemini-3.5-flash`. Flagship: `gpt-5.4` + `gemini-3.1-pro-preview`.
The pooled IG/HY flip exceeds both per-provider figures because pooling adds cross-provider disagreement absent within a single provider (Section 3.3).
Google flagship is one minor generation behind its flash counterpart (no Gemini-3.5 Pro exists).

3.8. Robustness across model tiers (post-hoc)

This analysis is *post-hoc*. It was not part of the pre-registered design and was carried out in a separate run after the confirmatory corpus was frozen, to address the concern that the instability might be an artifact of the low-cost model tier. We re-ran the frozen 12-specification grid over the 45 credit-health profiles at three seeds using each provider's flagship model (`gpt-5.4` and `gemini-3.1-pro-preview`), and compared it to the nano/flash tier evaluated on the identical grid. The per-comparison IG/HY flip is 23.6% (95% CI 19.4–27.8%) on the flagship tier versus 24.7% (95% CI 19.1–30.6%) on nano/flash; the paired difference is −1.1 percentage points (95% CI −7.6 to +6.0), not distinguishable from zero (Table 6). The three-way band flip is likewise statistically indistinguishable (−5.3 points, 95% CI −11.9 to +1.3). Flagship models are notably more lenient, they assign investment grade to 79.9% of elicitations versus 58.6% at the nano/flash tier, yet are no more consistent across elicitation choices. We do not detect attenuation of specification sensitivity at the flagship tier. The paired difference is small and statistically indistinguishable from zero, with a wide interval (−1.1 pp, 95% CI −7.6 to +6.0). This is an absence-of-evidence result, not evidence of absence, and is consistent with the instability being a property of how the categorical judgment is elicited rather than of model scale.

Two caveats attach to this comparison. First, the pooled figure masks provider heterogeneity. The flagship IG/HY flip is 26.7% for OpenAI (versus 17.3% at the nano tier) but 16.7% for Google (versus 20.5%), so the pooled near-equality reflects partial cancellation across providers rather than uniform stability. Second, the tiers are not perfectly generation-matched on the Google side, no Gemini-3.5 Pro model exists, so the flagship `gemini-3.1-pro-preview` is one minor generation behind its `gemini-3.5-flash` nano counterpart, whereas the OpenAI pair (`gpt-5.4` versus `gpt-5.4-nano`) is generation-matched. Finally, the flagship models do not recover the built Altman labels more often despite being easier to meet: flagship design-label recovery is 36.7% compared to 44.3% for nano/flash, so the 21-point investment-grade leniency shift reflects a more permissive rating posture, less close correspondence with the constructed labels.

3.9. Model class robustness: a reasoning architecture (post-hoc)

This analysis is *post-hoc*. It was not pre-registered and was collected in a separate run to address the worry that instability could be resolved by an explicit reasoning pipeline. We ran the frozen 12-specification grid over the 45 credit-health profiles at three seeds on an open-weight model with a native chain-of-thought, DeepSeek-R1 (via the OpenRouter API, frozen under `MANIFEST_REASONING.sha256`), adding a third model family and reasoning architecture. Because a single model instantiates the grid, this arm exercises the three non-provider axes (paraphrase, output format, presentation) and holds the fourth grid axis, provider, constant.

The per-comparison IG/HY flip is 25.5% (95% CI 20.9–30.1%), and the paired difference against the nano/flash tier is +0.9 percentage points (95% CI −7.0 to +8.8), indistinguishable from zero (Table 7). Two

Table 7
Reasoning-architecture arm (post-hoc, not pre-registered): per-comparison IG/HY flip on the frozen 12-specification grid (45 credit-health items, three seeds), and two-provider tiers for reference. The reasoning arm holds provider constant (single model). The paired difference is item-wise.

Tier	IG/HY flip (95% CI)	IG rate
Nano/flash (two providers)	24.7% (19.1–30.6)	58.6%
Flagship (two providers)	23.6% (19.4–27.8)	79.9%
Reasoning `deepseek-r1` (single model)	25.5% (20.9–30.1)	31.4%
Paired difference, reasoning − nano/flash: +0.9 percentage points (95% CI −7.0, +8.8). Reasoning arm varies three non-provider axes, provider is held constant.		

design differences qualify the comparison rather than make it strictly like-for-like. The reasoning arm shows no cross-provider variation (the largest single agreement gap, Section 3.3), and DeepSeek-R1 has a much more skewed marginal, giving investment grade to only 31.4% of elicitations, compared with 79.9% for the flagship tier. We therefore read the 25.5% flip as evidence of specification sensitivity within the evaluated reasoning model. In this arm, we do not detect attenuation of elicitation sensitivity under chain-of-thought reasoning.

3.10. Self-consistency: does aggregation remove the instability? (post-hoc)

This analysis is post-hoc and was not part of the pre-registration. A natural remedy is self-consistency. Instead of acting on a single elicitation, aggregate k elicitation observations and act on the majority vote. We quantify how much this helps by computing, for each issuer, the residual probability that two independent k-of-n majority votes land on opposite sides of the IG/HY boundary, averaged over issuers, for $k = 1, 3, 5, 9, 15$ (Fig. 4). Aggregation reduces but does not remove the instability. The residual flip falls from 25.2% at $k = 1$ to 19.4% ($k = 3$), 16.6% ($k = 5$), 13.6% ($k = 9$), and 11.1% at $k = 15$. Even fifteen aggregated elicitations therefore leave roughly one in nine matched issuer assessments crossing the investment-grade boundary. Self-consistency is a partial control whose cost grows linearly in k. It motivates reporting the residual dispersion explicitly rather than assuming a single call is representative, which is the role of the specification-instability score developed next.

4. A specification-instability dispersion index and review rule

Because the instability is a property of the elicitation process, it can be measured at inference time from the model's own outputs and reported alongside every rating. This section defines a descriptive dispersion index, benchmarks it against simpler measures of disagreement, and specifies the cost and governance use of a review rule based on disagreement. It does not validate an operational control, and Section 4.3 shows the index carries no incremental discrimination over counting dissenting elicitations at the resolutions tested.

4.1. Definition

For an issuer x, fix a plan of k elicitation observations drawn from defensible specifications, and a decision resolution g (for example, the map from letter grade to the IG/HY class). For each of the k planned elicitation observations, query the model under its assigned specification, map the resulting rating to its decision class, and let $p_{x,c}$ denote the resulting class proportion. The *specification-instability score* is the Gini–Simpson dispersion of that distribution,

$$\mathrm{SIS}(x) \;=\; 1 - \sum_{c} p_{x,c}^{2}, \tag{1}$$

reported alongside the modal rating $\hat{r}(x) = \arg\max_c p_{x,c}$. A score of zero denotes an invariant decision class at the chosen resolution across the evaluated elicitations. Higher values indicate greater sensitivity of the decision class to elicitation choices (the theoretical maximum for a binary decision is 0.5). The score requires no ground truth and no access to model internals, so it applies to any API-served model (Algorithm 1,

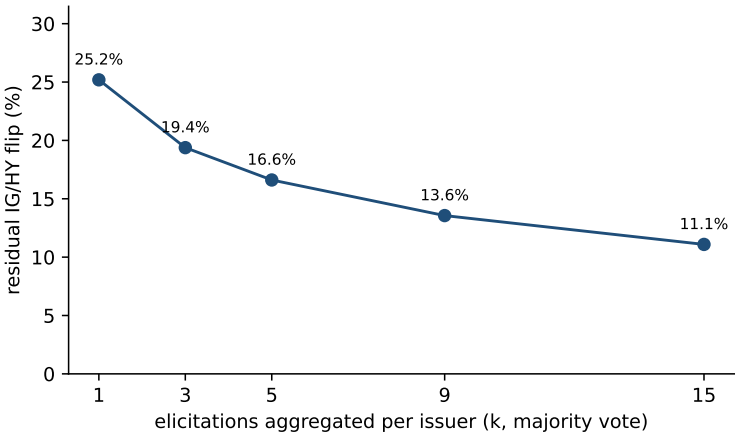


Figure 4: Self-consistency curve: residual per-comparison IG/HY flip after aggregating k elicitations per issuer by majority vote. Aggregation lowers instability with diminishing returns and does not eliminate it, about 11% of matched assessments still cross the boundary at $k = 15$. The curve is analytic rather than sampled. For each profile, p is its empirical share of investment-grade elicitations over all of its n_i elicitations (96 for 41 profiles, 95 for four). The k votes are modelled as independent Bernoulli(p) draws, so $q = \Pr(k\text{-vote returns IG})$ follows in closed form from the binomial, with ties at even k resolved to either class with probability one half. The plotted residual is the mean over profiles of $2q(1-q)$, the probability that two independent k-vote assessments of the same profile disagree.

Fig. 5). The Gini–Simpson form itself is a standard dispersion index. The contribution here is not a new mathematical index, and Section 4.3 shows it is ordinally equivalent to counting dissents at the resolutions tested. What the section offers is the held-out-specification evidence that early disagreement predicts later disagreement, a governance trigger built on it, and a cost model for running it.

4.2. Post-hoc validation on held-out specifications

This validation is *post-hoc* (it was not part of the pre-registration) and tests generalization across specifications, not across an external issuer sample. To test whether early disagreement is informative about later disagreement, we split the 32 specifications into a 16-specification calibration set and a disjoint 16-specification validation set, compute SIS for each issuer on the calibration set only, and ask whether it predicts that issuer's behaviour on the held-out validation specifications. The calibration score is strongly rank-correlated with the held-out flip rate (Spearman $\rho = 0.89$). Treating "the issuer's modal IG/HY call flips on at least one held-out specification" as the event to be flagged (base rate 0.76), the calibration score has a ROC-AUC of 0.94 (profile-clustered bootstrap 95% CI 0.86–0.99) and average precision of 0.98. To rule out sensitivity to a single partition, we repeat the analysis over 300 balanced random calibration/validation splits. The median AUC is 0.88 (interquartile range 0.86–0.91, full range 0.73–0.98). Calibration-set disagreement therefore identifies profiles that are also unstable on held-out specifications, under specifications not yet tried. Section 4.3 shows that this discrimination is not attributable to the index itself, because counting dissenting elicitations does as well, so the result is reported as evidence that *disagreement observed on one subset of specifications predicts disagreement on a disjoint subset*, not as validation of a new instrument.

We also repeat the calculation on the 31 perturbed real issuers as an illustrative descriptive real-issuer check only (Section 3.6), for which the same 12-specification grid was collected. Splitting those specifications 6/6, the calibration-set score gives ROC-AUC 0.70 (issuer-clustered bootstrap 95% CI 0.51–0.87, Spearman $\rho = 0.41$, base rate 0.61). This is descriptive only and is not evidence of generalization to real issuers, because residual ratio-pattern identification remains in this arm (Section 3.7). Discrimination is weaker than on the synthetic battery and the interval is wide, expected given only 31 issuers and a coarser six-specification calibration. We also report the recalibration requirement for a new model directly: cross-model transfer of the score is moderate but inconsistent across model pairs (per-item Spearman between one model's score and another's ranges from 0.14 to 0.73, median ≈ 0.45, and ≈ 0.55 across provider families, the AUC of

Algorithm 1. Dispersion index and review rule

input: issuer x. Elicitation plan yielding k observations. Resolution map g. Model M; letter scale $\mathcal{L}$ with boundary index b. Width w. Dissent threshold c_{ref}

$C \leftarrow [\,]$; $L \leftarrow [\,]$
for $j = 1, \ldots, k$:
  $\ell \leftarrow M(x, s_j)$ # one elicitation, on the letter scale
  append ℓ to L; append $g(\ell)$ to C # keep the letter and its class
$p_c \leftarrow |\{i : C_i = c\}|/k$ for each class c
$\hat{r} \leftarrow \arg\max_c p_c$; $d \leftarrow k - \max_c |\{i : C_i = c\}|$ # dissent count
SIS $\leftarrow 1 - \sum_c p_c^2$ # descriptive dispersion index
$\hat{\ell} \leftarrow \text{mode}(L)$ # modal *letter*, not the modal class
refer $\leftarrow (d \geq c_{\text{ref}})$ # the operational rule
priority $\leftarrow \left(|\text{rank}_{\mathcal{L}}(\hat{\ell}) - b| \leq w\right)$ # severity, not a gate
return $(\hat{r},\ \hat{\ell},\ d,\ \text{SIS},\ \textit{refer},\ \textit{priority})$

Figure 5: Dispersion index and review rule. Computable from any model served via API without access to model internals. Here k is the number of elicitation *observations* entering the calculation, which need not equal the number of distinct specifications: a one-call-per-specification workflow has $k = |S|$, whereas the retrospective validation of Section 4.2 retains three seeds per specification and so has $k = 16 \times 3 = 48$. The operational quantity is the integer dissent count d, referred when $d \geq c_{\text{ref}}$ for a governance threshold c_{ref} set by referral cost and capacity. Proximity to the boundary is returned as a severity indicator rather than a mandatory conjunct, because conjoining it with dissent cuts recall to 0.176 in our data. The near-boundary test is evaluated on the modal *letter* rating $\hat{\ell}$ and not on the modal decision class $\hat{r}$: at the investment-grade/high-yield resolution $\hat{r}$ takes only two values, so no subset of its range can express proximity to the boundary. With $\mathcal{L}$ the 21-grade letter scale, the boundary index is placed on a half-step convention, $b = \{\text{rank}_{\mathcal{L}}(\text{BBB}-) + \text{rank}_{\mathcal{L}}(\text{BB}+)\}/2$, so that $w = 1$ admits exactly the two grades straddling the cut, $\{\text{BBB}-, \text{BB}+\}$, and $w = 2$ additionally admits BBB and BB. This is an *operational near-boundary band* and is not the Altman Z″ WATCH stratum, which spans BB+ through B− and is a different construct. Thresholds on the continuous index would be k-dependent, which is one reason the operational rule is stated directly on the dissent count d.

one model's score for flagging another's above-median-instability items has median ≈ 0.72). Because this transfer is too variable to rely on, a new model should be re-scored on the grid rather than reusing another model's scores, at about \$7.43 per 1,000 issuers at $k = 9$ on a nano/flash tier (Table 10).

We state the operating rule in terms of the integer dissent count d rather than a cutoff on the continuous index, since Section 4.3 shows the index carries no information beyond d at these resolutions. The rule is $d \geq c_{\text{ref}}$, with the governance threshold c_{ref} chosen according to referral cost and capacity. "Refer on any disagreement" is the special case $c_{\text{ref}} = 1$, and we keep it only as a benchmark, because it is not the best choice once review is costly. On the held-out split $c_{\text{ref}} = 1$ refers 88.9% of profiles at precision 0.825 and recall 0.971, whereas $c_{\text{ref}} = 3$ refers 66.7% at precision 1.000 and recall 0.882, and the net benefit of $c_{\text{ref}} = 3$ is 0.6667, flat in the threshold probability, against 0.5000 for $c_{\text{ref}} = 1$ at $p_t = 0.60$ (Table 9). These are descriptive operating points on a split that also selected the operating point, not evidence of a validated instrument.

The number of observations matters for how any threshold reads, and k counts elicitation observations rather than distinct specifications. The retrospective validation retains three seeds per specification, so its calibration set has $k = 16 \times 3 = 48$, whereas a practical one-call-per-specification workflow has $k = |S|$. Thresholds on the continuous index are therefore k-dependent in a way that $d \geq c_{\text{ref}}$ is not, which is the second reason for preferring the count. Reported for completeness, the earlier Youden-selected cutoff $\tau = 0.08$ corresponds at $k = 48$ to $d \geq 3$, while at $k = 16$ it would collapse to $c_{\text{ref}} = 1$, since the smallest non-zero index value is 0.0408 at $k = 48$ and 0.117 at $k = 16$.

Proximity to the boundary is reported as a severity indicator rather than a mandatory conjunct. Instability is not confined to the boundary region, and requiring both conditions would cut recall to 0.176 while referring only 13.3% of profiles, trading referral volume against missed instability. Disagreement therefore determines whether a rating is referred, and proximity to the BBB−/BB+ cut determines how urgently. The continuous index remains useful for ranking within the referred set, and its discrimination (ROC-AUC 0.94) is scored on the full index rather than the binary gate, so at finer resolutions it takes many values rather than collapsing to a unanimity flag. That figure describes the retrospective $k = 48$

construction and is not evidence about the $k \approx 9$ aggregation level, which comes from the separate cost analysis of Section 4.4.

4.3. Benchmark against simple baselines

To assess whether the dispersion index adds information beyond simpler alternatives, we benchmark it against three baselines computable from the same k observations: the number of dissenting elicitations, the self-consistency vote margin, and the modal letter rating's distance to the investment-grade/high-yield boundary. Two further candidates are semantic entropy and token-level confidence. We compute the first and explain why the second cannot be computed. Farquhar, Kossen, Kuhn and Gal (2024) give a *discrete* semantic entropy estimator that approximates the probability of each semantic cluster from the number of generations falling in it, disregarding token probabilities, and recommend it precisely for models that do not expose probabilities. Our outputs are categorical ratings on a fixed ordinal scale, so the semantic-equivalence clustering those authors obtain by bidirectional entailment over free text is given here by construction, because two elicitations share a cluster exactly when they share a rating at the resolution analysed. Discrete semantic entropy then reduces to the Shannon entropy $H = -\sum_c p_c \log p_c$ of the rating distribution, which the frozen corpus does support, and we report it as a fifth predictor. Two scope limits must be stated. First, Farquhar et al. (2024) draw repeated generations under a *fixed* prompt to detect confabulation, whereas our ensemble deliberately varies the elicitation specification. What we compute is therefore a specification-level analogue of their estimator applied to a categorical outcome, not a replication of their hallucination-detection task. Second, token-level confidence genuinely *cannot* be computed, because the corpus retains parsed text and no log-probabilities, and obtaining them would require re-elicitation with log-probabilities enabled, which is not uniformly available across the providers evaluated.

We run the comparison in both the held-out-specification setting and a profile-split subsampling analysis, and report them separately in Table 8 rather than merging them. The profile-split analysis splits the 45 credit-health profiles at random into halves over 300 draws and evaluates discrimination only on the held-out half. The conclusion is identical, with mean AUC 0.936 at the gate and 0.950 at three bands and an index-minus-dissent-count difference of exactly zero on every split. At the binary gate this is expected rather than reassuring, because the equivalence there is algebraic and so cannot depend on which profiles are evaluated. At three bands the measures are not rank-identical and the agreement in AUC is an empirical finding of these data rather than an identity.

The comparison is unfavourable to the score, and for a reason that is algebraic rather than empirical. At a binary resolution with k elicitations and minority count m, the index equals $1 - \left[(m/k)^2 + (1-m/k)^2\right] = 2(m/k)(1-m/k)$, which is strictly increasing in m for $m < k/2$. The score is therefore an order-isomorphic transform of the dissent count at the binary gate, and any rank-based discrimination measure must give the two identical values there. The same argument covers discrete semantic entropy, since the binary entropy function $H_b(m/k)$ is also strictly increasing on $m < k/2$, so at the decision gate the dispersion measure one chooses is immaterial. The scope of this statement matters. At the binary gate the equivalence is algebraic. At three bands the measures are *not* rank-identical, with a Spearman correlation of 0.971 between the index and the dissent count, yet they yield the same AUC in the present specification-split and profile-split analyses. Table 8 confirms this numerically and extends it: at the three-band resolution the two agree to within numerical precision as well, and at the full letter resolution the exercise is vacuous because every item changes letter on some held-out specification, so the event to be predicted has a base rate of one.

Two conclusions follow. First, the index adds no discrimination over counting dissents, so the operating rule can be stated without it, as $d \geq c_{\text{ref}}$ on the integer dissent count. The choice of c_{ref} is a governance decision rather than an empirical one, and the decision curve shows that a stricter c_{ref} dominates $c_{\text{ref}} = 1$ once review is costly. The continuous score contributes nothing beyond the count at the decision resolutions a credit gate actually uses. Second, distance to the boundary is a genuinely *different* and weaker signal (AUC 0.763 and 0.655), so it is not a substitute for the dissent count. The two measure different things, which is why Algorithm 1 (Fig. 5) uses dissent to determine referral and boundary proximity to determine priority. We therefore describe the index as a descriptive dispersion summary and a convenient governance trigger, and not as a control with incremental value over simple disagreement counting.

Table 8
Discrimination of the instability index against baselines computable from the same k calibration elicitations, predicting whether an issuer's call changes on at least one held-out specification. The upper block uses the 16/16 specification split over all 45 credit-health profiles. The lower block repeats the comparison in the profile-split analysis, splitting those profiles at random into halves over 300 draws and scoring only the held-out half. Throughout this section the unit is a constructed credit-health profile, not one of the 31 real issuers of Section 3.6. AUC uses mid-rank tie correction, which the heavily tied baselines require. Discrete semantic (Shannon) entropy is the estimator of Farquhar et al. (2024) applied to our categorical outcome, for which semantic clusters are the rating categories. Letter-resolution AUC is undefined because every profile changes letter on some held-out specification, so the event base rate is one.

Predictor	IG/HY	3 bands	letter
Instability index (Gini–Simpson)	0.936	0.951	—
Discrete semantic (Shannon) entropy	0.936	0.951	—
Number of dissenting elicitations	0.936	0.951	—
Self-consistency vote margin	0.936	0.951	—
Distance to IG/HY boundary	0.763	0.655	—
Event base rate	0.76	0.82	1.00
Spearman (index, dissent count)	1.000	0.971	0.938
Spearman (entropy, dissent count)	1.000	0.890	0.837
AUC difference, index − dissent count	0.000	0.000	—
95% CI (issuer-clustered, 2,000 draws)	(0.000, 0.000)	(0.000, 0.000)	—
AUC difference, entropy − dissent count	0.000	0.000	—
Profile-split subsampling (300 random splits of the 45 profiles)			
Instability index (Gini–Simpson)	0.936	0.950	—
Discrete semantic (Shannon) entropy	0.936	0.950	—
Number of dissenting elicitations	0.936	0.950	—
Self-consistency vote margin	0.936	0.950	—
Distance to IG/HY boundary	0.768	0.649	—
AUC difference, index − dissent count	0.000	0.000	—
maximum |difference| across splits	0.000	0.000	—

Net benefit for triage. Discrimination is not the quantity a referral desk optimizes, so we also report the decision-curve net benefit of Vickers and Elkin (2006), $\mathrm{NB}(p_t) = \mathrm{TP}/n - (\mathrm{FP}/n)\, p_t/(1-p_t)$, where p_t is the threshold probability at which a reviewer is indifferent between referring and not referring. Each continuous predictor is credited with the best net benefit attainable over its *own* thresholds, which is the most favourable reading available to it. Table 9 gives the curves. Three results follow, and the first two are null.

The index and the dissent count have *identical* net-benefit curves at every threshold probability, to machine precision, at both resolutions where the event is non-degenerate. The largest absolute gap is exactly 0. This is the order-isomorphism again, now in the decision metric. The two predictors' threshold families generate the same flag sets, so no triage policy can distinguish them. The self-consistency vote margin coincides with them as well. Distance to the boundary is dominated, by up to 0.156 in net benefit. The index therefore yields no triage gain over counting dissents, and we report this null as the answer to the question.

Dispersion-based referral does, however, beat the two reference policies once referral is costly enough. Because the event base rate is high (0.76 at the binary gate), referring every issuer is already near-optimal for $p_t \leq 0.20$, where the gain from any predictor is 0. From $p_t = 0.25$ the dispersion predictors separate, reaching $+0.278$ over refer-all at $p_t = 0.60$. The useful knob there is the *number* of dissenting elicitations rather than the continuous score. The fixed "refer on any disagreement" rule falls short of the attainable envelope by up to 0.167, because for $p_t > 0.30$ a count threshold above one dissent is preferable. Because the operating parameter is an integer count on a grid of $k+1$ attainable values rather than a tuned continuous cutoff, no threshold selection is required, and we state the count rule and its net benefit directly.

Table 9
Decision-curve net benefit for triage, $\mathrm{NB}(p_t) = \mathrm{TP}/n - (\mathrm{FP}/n)\, p_t/(1-p_t)$, 45 credit-health profiles, same 16/16 specification split. Continuous predictors are credited with the best value attainable over their own thresholds. The letter resolution is omitted, because its event base rate is 1.00, so refer-all is optimal at every p_t and the curve is degenerate. The analysis specification was committed and hashed before the computation was run (`SPEC_r3_decision_curve.md`, deposited with the archive).

	threshold probability p_t				
Strategy (IG/HY resolution)	0.10	0.20	0.30	0.45	0.60
Instability index (own best threshold)	0.728	0.694	0.667	0.667	0.667
Discrete semantic (Shannon) entropy	0.728	0.694	0.667	0.667	0.667
Number of dissenting elicitations	0.728	0.694	0.667	0.667	0.667
Self-consistency vote margin	0.728	0.694	0.667	0.667	0.667
Distance to IG/HY boundary	0.728	0.694	0.651	0.572	0.511
"Refer on any disagreement" (fixed)	0.716	0.694	0.667	0.606	0.500
Refer all issuers	0.728	0.694	0.651	0.556	0.389
Refer none	0.000	0.000	0.000	0.000	0.000
Maximum absolute gap across the full p_t grid, 0.05–0.60:					
index − dissent count	0.000 (exact)				
entropy − dissent count	0.000 (exact)				
index − vote margin	0.000 (exact)				
index − boundary distance	0.156				
attainable envelope − "any disagreement"	0.167				

Table 10
Cost of repeated elicitation by aggregation level k: cost to score 1,000 issuers, from the observed per-call token counts and provider pricing.

k	Tokens/issuer	$/1,000 (nano/flash)	$/1,000 (flagship)
1	416	0.83	13.12
3	1,248	2.48	39.37
5	2,080	4.13	65.61
9	3,745	7.43	118.10
15	6,242	12.38	196.84

Token counts are the observed per-call means. Dollar figures use provider list pricing accessed August 2026 from the OpenAI and Google API pricing pages and will change as pricing changes.

4.4. Aggregation level and computational cost

The number of elicitations k trades stability against cost. The self-consistency curve (Section 3.10) shows steeply diminishing returns. The modal call's residual IG/HY flip falls from 25.2% at $k = 1$ to 13.6% at $k = 9$ and 11.1% at $k = 15$, so a moderate $k \approx 9$ captures most of the achievable stabilization while still exposing residual dispersion through the score. The cost is modest and linear in k: at $k = 9$, scoring one thousand issuers requires about 3.7 million tokens and costs roughly $7 on a nano/flash tier or $118 on a flagship tier (Table 10). Because the score is computed from calls the workflow already makes, its marginal cost over simply issuing a rating is the $k - 1$ additional elicitations.

4.5. Governance workflow

In a model-risk workflow the score sits between the model and the decision (Fig. 6). Each issuer is elicited k times, and the modal rating and dissent count are recorded. Issuers whose dissent count reaches the governance threshold c_{ref} are routed to human review, prioritized by proximity of the modal letter rating to the BBB−/BB+ cut. Issuers below the dissent threshold pass through. This yields an auditable

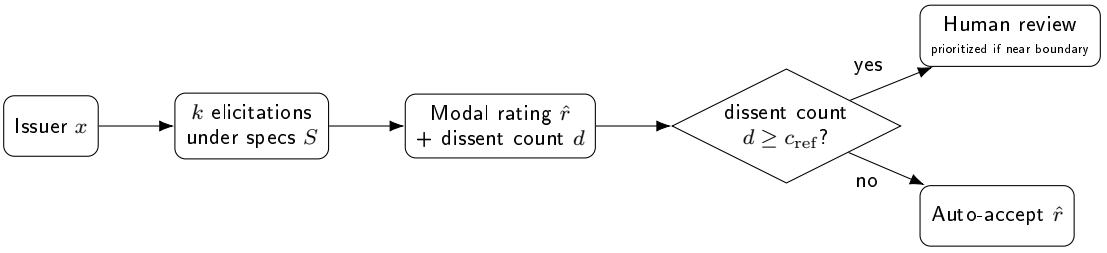


Figure 6: Governance workflow. Issuers whose dissent count reaches the governance threshold c_{ref} are routed to human review, prioritized by proximity of the modal letter rating to the BBB−/BB+ cut. Issuers below the dissent threshold pass through.

review rule, a governance trigger that scales with the rating pipeline and speaks to a supervisory model-risk question: how much of this rating is a property of the firm, and how much of how we asked?

5. Discussion

We evaluate whether an LLM-generated credit rating can be treated as a stable assessment of a firm's financial health or whether it depends heavily on how that measure is elicited. The main finding of the pre-registered specification curve is that substantial instability remains even after controlling for the underlying financial information. As expected, firm characteristics explain most of the total variation in ratings, but the residual variation due to elicitation is large enough to change economically meaningful decisions.

The variance decomposition clarifies the origin of this instability, and its interpretation is important. Under the fitted models the elicitation axes jointly explain $R^2 = 0.083$–0.119 of rating variation, and the design does not identify how that share should be attributed to individual axes. Of the ∼25% OLS residual, only about 8% is pure run-to-run seed variation, and the rest is not identified as to source under this resolution-limited design and is reported descriptively. The headline stability conclusions do not depend on this decomposition. The instability is nonetheless not dismissible as mere sampling noise (pure-seed variance is a minority of the residual), and the headline flip and agreement results are model-free.

It is useful to separate two kinds of risk this induces, because they call for different controls. *Build-time* risk is the one-time consequence of fixing a particular provider, model snapshot, and prompt template when a system is designed. The elicitation axes *jointly* explain only a small and model-dependent share of the variance (R^2 between 0.083 and 0.119 across the specifications we fit), no individual axis share is identified under this design (Section 6.2), and the one separately identified interaction, provider×item at 1.4%, is non-negligible; the chosen configuration therefore selects a point on the specification curve that need not generalize to other defensible configurations. *Run-time* risk is the ongoing 8.1% pure seed variation that recurs on every call even with the configuration held fixed. The two map to different mitigations, build-time risk is managed by evaluating across specifications before deployment (the specification-instability score of Section 4), run-time risk by aggregating repeated calls (self-consistency, Section 3.10), and to different lines of a supervisory model-risk framework such as SR 11-7, one concerning model selection and validation, the other ongoing monitoring.

The real-issuer experiment complements, rather than substitutes for, this identification strategy. Crucially, it first demonstrates that direct evaluation on anonymized real firms is not automatically clean, because a model identifies around 22% of the unperturbed issuers from financial information alone. After issuer-specific scaling and screening, the fingerprint rate falls to 2.9%, and substantial specification variation is still observed. The real-issuer arm is therefore treated as an illustrative descriptive check. Its sample size does not support quantitative equivalence with the controlled battery.

5.1. Mitigating exact-figure recognition in real-issuer evaluations

The fingerprinting result also has implications beyond the credit-rating experiment at hand. Public companies' financial information is widely disseminated and may have been included in the training data of general-purpose LLMs. Removing company names does not necessarily remove issuer identity from an

evaluation: if a model can infer the issuer from the precise financial numbers, then a judgment that appears to be based on the inputs may also draw on memorized information about that company. Perturbation combined with explicit identification testing can reduce exact-figure recognition, but Section 3.7 shows it is not sufficient to establish decontamination. A ratio-preserving perturbation leaves the ratio pattern intact, and ratio-pattern identification must therefore be assessed separately before an evaluation on public financial entities is treated as contamination-free.

Taken together, the results suggest that a credit rating produced by an LLM should not be interpreted as a specification-invariant point estimate. Each response is one realization from a broader distribution induced by the elicitation procedure. This does not imply that LLMs cannot contribute to credit analysis, and the present study does not assess whether their ratings are more or less accurate than established credit-assessment methods. Rather, a prerequisite for reliable use is to characterize the uncertainty associated with reasonable elicitation choices before treating a point rating as a stable model output.

5.2. Limitations

These findings have several limitations that bound their scope. First, the study considers two model providers and the specific model snapshots and configurations included in the pre-registered design (Table 12). The results should therefore be viewed as evidence of how the evaluated systems behave, not as a universal numerical estimate for all current or future LLMs. Relatedly, the evaluated snapshots are each provider's low-cost (`nano`/`flash`) tier rather than a frontier model. We study this tier deliberately, because cost pressure makes it the realistic choice for high-volume, per-issuer credit screening. The concern that specification sensitivity might attenuate at the frontier is examined in Section 3.8: on the same grid, flagship models (`gpt-5.4`, `gemini-3.1-pro-preview`) show an IG/HY flip statistically indistinguishable from the nano/flash tier. We interpret this as not detecting attenuation (wide interval, absence of evidence result) rather than as proof that scale does not matter. The reasoning-architecture arm (Section 3.9) also shows specification sensitivity remains in another model class. The flagship comparison is based on one model per provider on a reduced grid, and a fuller frontier sweep remains a useful extension.

Second, the main identification sample is a fixed 90-item constructed battery, with 45 profiles in the credit-health task and 45 in the directional task. Repeated specifications increase coverage of the elicitation space but not the number of independent underlying firm profiles, so the flip rates and variance estimates remain sample statistics for the 45 credit-health items considered. We report profile-clustered confidence intervals throughout to make this uncertainty explicit.

Third, the real-firm arm comprises 31 U.S. mid-cap business-to-business issuers selected under the perturbation-and-fingerprint-screening design. This sample is an illustrative descriptive real-issuer check and is much smaller than the controlled battery at the relevant strata. The WATCH-stratum sample is small, and quantitative equivalence between the real and constructed estimates is not established.

Fourth, the study measures stability, not absolute credit-rating capability. The Altman Z'' mapping provides a pre-specified accounting-based benchmark for constructing and evaluating the controlled credit-health battery, while disclosed agency ratings provide an alternative descriptive reference. Neither is treated as a unique ground-truth credit opinion. Whether generative models can produce more accurate credit assessments than established statistical, market-based, or agency methods is a separate research question.

A further methodological caveat concerns the variance decomposition. The 32-specification grid is a balance-constrained D-optimal fraction chosen to make the main effects mutually near-orthogonal, but with N not divisible by the strength-2 requirement the design is resolution-limited: main effects can be aliased with two-factor interactions. The overlap is substantial and precisely characterised: the main-effect and two-factor-interaction subspaces have a four-dimensional exact intersection, so some linear combination of main effects coincides exactly with a combination of interactions, while no individual axis contrast is itself exactly confounded, each lying in the interaction span with R^2 between 0.690 and 0.901 (Section 6.2). We therefore read the decomposition main-effects-first and report the structured portion of the residual as "unmodeled interactions" rather than attributing it to specific axes. The headline stability results (flip shares, agreement) are model-free and do not depend on the ANOVA partition.

Perturbation plus exact-figure fingerprint screening can reduce exact-number recognition but is insufficient to establish decontamination, and ratio-pattern identification must also be assessed. Within these bounds, the main stability finding is robust across the specification curve, the temperature-0-honored subgrid,

the provider comparison, and the rating-resolution analysis, and is further illustrated, descriptively only, by the perturbed real-firm arm, which Section 3.7 shows is not decontaminated enough to support an external-validity claim.

5.3. Future directions

The present design suggests several extensions. Larger and sector-stratified real-issuer samples would yield more precise estimates of how specification instability varies with firm characteristics and distance from rating boundaries. Similar designs could test retrieval-augmented or tool-using credit agents, where access to external information may reduce some forms of uncertainty while adding further specification choices. The proposed specification-instability score could be calibrated against downstream decision quality and integrated into human-in-the-loop workflows. More broadly, the perturbation-and-fingerprint-screening procedure could be applied to other evaluations of financial LLMs involving publicly documented entities: explicitly testing whether an ostensibly anonymized case remains identifiable, in tasks such as risk assessment, valuation, investment recommendation, and supervisory analysis, may help separate independent reasoning from memorized entity knowledge.

Overall, reproducibility in AI-assisted credit assessment cannot be taken for granted from a single elicitation. We present these findings as a case study of the evaluated systems rather than as a universal numerical constant. Across a pre-registered specification curve, firm fundamentals explain most of the total variation in machine-generated ratings, but substantial specification-level instability remains after controlling for the underlying financial information, only a minority of which is pure run-to-run noise. For the systems studied here, the elicitation procedure is part of the measurement: unless stability with respect to reasonable elicitation choices has been demonstrated, a machine-generated credit rating should not be treated as a specification-independent point estimate, and the specification-instability score offers a concrete way to report that uncertainty.

6. Methods

6.1. Design, battery, and benchmark

The study is designed to measure *elicitation stability*, meaning whether an LLM produces a credit assessment that is consistent when the underlying financial information is held constant but reasonable choices in how the assessment is elicited vary. The aim is therefore not to estimate rating accuracy on a naturally occurring population of firms, but to isolate variation attributable to the elicitation process.

We use a constructed battery as the main identification environment because it allows firm fundamentals to be held fixed while the elicitation specification varies, permits balanced representation across credit-health conditions, and provides a benchmark that can be computed before any model is queried. Restricting the analysis to public companies would introduce a further source of uncertainty, because an LLM might identify an issuer from financial information it was exposed to during training. We therefore use the controlled battery for the main analysis and a separately perturbed real-firm sample for a complementary illustrative descriptive real-issuer check.

The battery consists of 90 fixed firm-quarter profiles. Each profile is summarized in a compact financial statement, including revenue, EBIT, total assets and liabilities, equity, working-capital figures, and retained earnings. The profiles comprise two equally sized task families: 45 credit-health items, for which the model produces a credit rating, and 45 directional items, for which the model produces a buy/hold/sell call. All analyses in this paper use the 45 credit-health profiles. The 45 directional profiles are part of the pre-registered battery but are not analyzed here.

For the credit-health task, each profile is associated with a pre-specified Altman Z'' accounting-based reference (Altman, 1968), computed with coefficients 6.56, 3.26, 6.72, and 1.05 and cutoffs 2.60 and 1.10 ($Z'' \geq 2.60$ = safe, $1.10 < Z'' < 2.60$ = watch, $Z'' \leq 1.10$ = distress). The resulting safe, watch, and distress strata each contain 15 observations, evenly distributed across the 45 credit-health profiles. The machine's letter rating is mapped deterministically to notch, three-band, and IG/HY outcomes as in Table 11. The Altman mapping provides a deterministic accounting reference for constructing and stratifying the controlled battery and is not treated as a unique ground-truth agency rating, because our interest is stability across elicitation specifications rather than absolute rating accuracy. The benchmark and battery were frozen under

Table 11
Deterministic mapping from the machine letter rating to coarser resolutions, and the Altman Z″ benchmark strata. The three-way band and the IG/HY split share the same investment-grade cut.

Letter grade	Notch	Three-way band	IG/HY
AAA	AAA	SAFE	Investment grade
AA+	AA	SAFE	Investment grade
AA	AA	SAFE	Investment grade
AA−	AA	SAFE	Investment grade
A+	A	SAFE	Investment grade
A	A	SAFE	Investment grade
A−	A	SAFE	Investment grade
BBB+	BBB	SAFE	Investment grade
BBB	BBB	SAFE	Investment grade
BBB−	BBB	SAFE	Investment grade
BB+	BB	WATCH	High yield
BB	BB	WATCH	High yield
BB−	BB	WATCH	High yield
B+	B	WATCH	High yield
B	B	WATCH	High yield
B−	B	WATCH	High yield
CCC+	CCC	DISTRESS	High yield
CCC	CCC	DISTRESS	High yield
CCC−	CCC	DISTRESS	High yield
CC	CC	DISTRESS	High yield
C	C	DISTRESS	High yield

The notch resolution drops the +/− modifiers, collapsing the 21 letter grades to 9 notches. *Altman Z″ strata:* SAFE ≥ 2.60; WATCH 1.10–2.60; DISTRESS ≤ 1.10.

a pre-registration hash before any model query, and an independent post-hoc re-execution from the frozen financial facts reproduces all 45 benchmark classifications.

6.2. Specification grid and data collection

We manipulate seven researcher-controlled elicitation dimensions: model provider (OpenAI, Google), model version (a current and a prior snapshot per provider), temperature (0 or 1.0), prompt paraphrase (three semantically equivalent formulations), output format (JSON or free-text), few-shot exemplars (zero or two), and answer presentation (original prose versus a table with reversed option ordering). The exact model snapshots and the collection window are listed in Table 12. A balance-constrained D-optimal fractional grid was generated from an effect-coded main-effects model with the design seed frozen in advance. The analyzed two-provider design comprises 32 specifications, each evaluated over three seeds. Provider metadata were preserved as returned, including effective generation settings where available. The requested temperature of 0 was not uniformly honored by providers.

The confirmatory experiment therefore comprises $90 \times 32 \times 3 = 8{,}640$ elicitation cells (4,320 on the 45 credit-health profiles), of which 8,633 returned substantive content. The seven non-substantive cells (empty or unparseable responses, 0.08%) divide as four on the credit-health profiles and three on the directional items, so the credit-rating analyses in this paper run on 4,316 of the 4,320 designed cells. The four absent cells are all on specification S23 and each is short one of three seeds. They were not re-collected, because the corpus is frozen and re-querying after the fact would break that. Where this paper reports 4,320 it is describing the design. Every statistic is computed on the 4,316. The non-substantive cells are dropped listwise. They enter no agreement, flip, or variance computation, and because every statistic is an average over the cells that did parse, their omission cannot manufacture instability. Each call is stored as an immutable JSON record containing the experimental specification, the returned content, and the available provider metadata. Data

Table 12
Model snapshots evaluated. The four confirmatory snapshots (two providers, current and prior each) constitute the main 8,640-cell experiment. The two flagship snapshots are the separate model-tier robustness arm (Section 3.8). Responses were collected once and frozen. All reported results reproduce offline from the frozen corpus.

Provider	Model snapshot	Role
OpenAI	`gpt-5.4-nano`	current (confirmatory)
OpenAI	`gpt-5-nano-2025-08-07`	prior (confirmatory)
Google	`gemini-3.5-flash`	current (confirmatory)
Google	`gemini-2.5-flash`	prior (confirmatory)
OpenAI	`gpt-5.4`	flagship (robustness)
Google	`gemini-3.1-pro-preview`	flagship (robustness)
DeepSeek	`deepseek-r1` (OpenRouter)	reasoning (robustness)

The four confirmatory snapshots were queried once in the frozen confirmatory run (July 2026). The two flagship snapshots in the later robustness run (August 2026), and the reasoning snapshot (`deepseek-r1`, served via the OpenRouter API) in a separate reasoning run (September 2026). All were used offline thereafter. Aliases without an embedded date stamp are pinned by these query months, with the exact dates recorded in the frozen manifests. The reasoning model is API-served, so its temperature and seed are request parameters, not self-hosted determinism.

collection and analysis are separated: all confirmatory analyses run only on the frozen response corpus and do not re-query any model.

What the design identifies. Write the effect-coded main-effects model as $y = X\beta + \varepsilon$ and let Z collect the omitted two-factor interaction contrasts. Then $\mathbb{E}[\hat{\beta}] = \beta + A\gamma$ with alias matrix $A = (X'X)^{-1}X'Z$, so the bias in any main effect j satisfies $|a_j'\gamma| \leq \|a_j\|_2 \, \|\gamma\|_2$. On the 32-specification confirmatory design there are 27 such interaction contrasts and the alias norms are substantial: $\|a_j\|_2$ ranges from 0.68 to 1.04 across the eight main-effect contrasts, and for answer presentation it is 0.878, its largest single partner being the version-by-few-shot interaction at $|a| = 0.484$. **No finite bound on a per-axis share therefore follows from the design alone**. The inequality is informative only once an external constraint is placed on $\|\gamma\|$, and nothing in the design supplies one. Since $\|a_j\|_2 \approx 0.9$ for presentation, omitted interactions jointly no larger than the largest estimable main effect would suffice to move that axis's apparent share by about as much as the share itself, which is why we report no per-axis ranking.

Both the alias norms and the pairwise column correlations depend on a coding convention, since effect coding is not invariant to which level is taken as the reference. We therefore add a diagnostic that is invariant to any non-singular re-parameterisation within either subspace. The canonical correlations between the centred main-effect subspace and the centred two-factor-interaction subspace. On the analysed design the largest is 1.000 and 6 of 8 exceed 0.9. A value of one means that some linear combination of main effects coincides *exactly* with a combination of two-factor interactions, and that is what occurs here. The joint matrix $[X \; Z]$ has rank 31 rather than 35, so the two subspaces intersect in 4 dimensions. We state the implication precisely, because the distinction matters. It is *not* the case that any single axis contrast is exactly confounded. Regressing each main-effect contrast on the full interaction span gives R^2 between 0.690 and 0.901. What is exactly confounded is a combination. That is sufficient for non-identification of the individual shares, and it cannot be dismissed as an artifact of how the contrasts were coded.

The provider-by-item interaction is in a different position and is identified up to a numerically negligible imbalance. Items are fully crossed with specifications: in the analysed arm all $45 \times 32 = 1{,}440$ item×specification cells are populated, 1436 of them with the full three seeds and 4 with two. Because no specification-axis contrast involves the item factor, crossing makes the provider-by-item contrasts orthogonal to every specification main effect and two-factor interaction up to that imbalance. The largest absolute correlation realised is 0.0018 against the main effects and 0.0022 against the interactions. We therefore report this share without an assumption about γ, while noting that the orthogonality is numerical rather than exact. The aliasing problem is confined to the specification axes among themselves.

6.3. Outcome construction and statistical analysis

Stored model responses are parsed under three rules, strict, lenient, and tolerant, applied after collection. Credit ratings are first expressed on the full 21-grade letter scale and then deterministically aggregated into notch categories, three broad credit bands, and the binary investment-grade/high-yield classification (Table 11). This permits examination of stability at increasingly coarse rating resolutions without additional model calls.

Agreement across specifications is measured with Fleiss' κ. Within- and cross-provider agreement is computed on matched firm profiles and seeds.

Matched-pair disagreement: sampling convention. Every reported "matched-pair" flip probability is the Gini–Simpson index of an item's own elicitations at the stated resolution, $1-\sum_c p_c^2$, where p_c is the share of that item's elicitations falling in class c. This is exactly the probability that *two draws taken independently and with replacement* from the item's elicitation set fall in different classes. Three properties follow and are stated explicitly because they determine reproducibility. First, the two draws are taken *with* replacement, so they may select the same elicitation cell. Such a pair trivially agrees and contributes to the agreement mass. Second, the draws are taken from the item's *pooled* elicitations, so a pair may differ in specification, in seed, or in both. The statistic is not conditioned on holding either fixed. Third, the item, and hence the underlying firm profile, is always held fixed. Pairs are never formed across items. In the confirmatory battery 41 profiles have $n_i = 96$ elicitations and four have 95. Throughout, k denotes the number of elicitation *observations* entering a disagreement or dispersion calculation, not the number of distinct specifications: the retrospective 16-specification validation retains three seeds and so has $k = 16\times3 = 48$, whereas a one-call-per-specification workflow has $k = |S|$. The without-replacement variant, which excludes same-cell pairs, is the same quantity multiplied by $n_i/(n_i-1)$, evaluated at each profile's own n_i (96 for 41 profiles, 95 for four), which raises the pooled estimate from 25.2% to 25.45%. Variation in the full letter-rating rank is decomposed by ordinary least squares with Type-II analysis of variance, with terms for the underlying item, the elicitation-design dimensions, the provider-by-item interaction, and a residual. To interpret the residual, we additionally estimate the *pure within-cell seed variance* as the sum of squares across the three seeds within each identical item×specification cell, divided by the total sum of squares. The difference between the OLS residual and this pure-seed component is reported as "other unexplained" (unmodeled interactions).

All analyses are pre-registered unless labelled post-hoc, and inference uses distribution-free resampling throughout, so no normality assumption is made. Uncertainty for the main quantities is estimated by profile-clustered percentile bootstrap resampling over the underlying firm profiles, instead of over individual model calls, thereby preserving dependence among repeated elicitations of the same profile. We report the variance shares (item and pure-seed) and the IG/HY flip rate with two-sided 95% confidence intervals (2,000 replicates for the flip rate, 1,000 for the variance-share and specification-curve quantities). Specification invariance under the temperature-0-honored setting is tested on the provider-setting combination whose metadata confirm that temperature was honored (Google at temperature 0). We test a rating-invariance null: specification-condition assignments are permuted across profiles (rather than permuting accuracy labels), and the statistic is the between-specification spread of the item-centred letter rating, compared with the distribution generated by 2,000 such permutations in a one-sided test, $p = (k{+}1)/(n{+}1)$, where k is the number of permuted statistics at least as extreme as observed and $n = 2{,}000$. The test is one-sided in that only an observed spread greater than the permutation distribution provides evidence against specification invariance. The real-firm arm is presented descriptively. The real-firm and constructed-battery WATCH-stratum flip shares are compared by their issuer-clustered bootstrap difference and interval, with no equivalence claim. The significance threshold is $\alpha = 0.05$ for all formal tests. To mitigate the risk of selective reporting within the specification multiverse, we pre-registered the estimands and specification grid and report the entire specification curve rather than a selected subset. We reserve "significant" for results accompanied by a p-value and otherwise use "substantial" or "considerable."

For the binary IG/HY outcome, the per-comparison flip probability is computed separately for each credit-health profile from its observed distribution of ratings across the entire elicitation grid, so that two elicitations may differ on any specification dimension or seed. The resulting pair falls on opposite sides of the IG/HY boundary with probability equal to one minus the Gini–Simpson concentration of the profile's binary labels, and the reported flip rate is the average of this quantity over the 45 credit-health profiles. The

same per-issuer quantity, reported at the decision resolution of interest, is the specification-instability score developed in Section 4.

Three post-hoc analyses are run offline from the frozen corpora. The specification-instability index is evaluated descriptively on a 16/16 held-out-specification split and benchmarked against simpler measures of elicitation disagreement. No operating threshold is proposed or validated. The range of variance explained by the elicitation axes is described in a main effects model, a main plus two-way interaction model, and an issuer-out-of-sample regularized model, and the real-firm external-benchmark specification curve is tested with an issuer-preserving permutation (specification labels permuted within issuer). The reasoning-architecture arm (Section 3.9) is frozen under `MANIFEST_REASONING.sha256`.

6.4. Real-firm arm

To test the persistence of specification instability outside the constructed battery, we assemble a complementary sample of 31 U.S. business-to-business issuers with market capitalizations between $2 billion and $15 billion whose most recent Form 10-K discloses a current long-term agency credit rating. S&P is the primary rating source. Where S&P ratings are unavailable, Moody's or Fitch ratings are mapped to a common scale. Each rating is preserved with a verbatim filing quotation, a permanent SEC document URL, and a document hash.

Because these figures are publicly reported by real issuers, there is a risk of contamination if a model can identify a company. We therefore first perform a pre-registered fingerprinting test in which the models are asked to identify issuers from anonymized financial profiles. Gemini-Flash identifies about 22% of the selected issuers from the original figures, confirming that removing the company name does not guarantee independence from memorized issuer information. To reduce this risk, every dollar-denominated quantity for an issuer is multiplied by a single issuer-specific factor sampled from a log-uniform distribution over $[0.6, 1.7]$. The same factor applies to all monetary amounts, so the credit-health analysis retains the financial-ratio structure while disrupting exact-figure lookup. We also exclude consumer-facing companies to reduce recognition from highly distinctive public profiles. The fingerprint-identification rate falls to 2.9% after perturbation, below the pre-registered 5% fingerprint-identification threshold.

Each retained perturbed issuer is then scored on a fixed 12-specification grid spanning presentation, provider, prompt paraphrase, and output format, at three seeds, and the constructed battery is re-scored under the same grid for a matched comparison on the WATCH stratum. This arm uses disclosed agency ratings as a descriptive real-world reference. The primary comparison is stability across specifications.

6.5. Reproducibility

The entire experimental pipeline is archived so that the reported results can be reproduced without further model calls. The frozen materials comprise the 90-item battery, the specification grid, the prompt templates, the rating scale, the frozen response corpus, the analysis panel, the real-firm arm, the analysis code, the generated outputs, and SHA-256 integrity manifests. The complete set of confirmatory tables, figures, and reported statistics regenerates from a single offline entry point. Provider API credentials are required only to collect new model responses and are not needed to reproduce any result reported in this study.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Generative AI was used to assist with grammar correction and minor language edits of the authors' own text, and with software scaffolding, development, and coding of the analysis and reproducibility pipeline (Anthropic Claude), in all cases under the authors' direction. Separately, large language models are the object of study in this work, the credit-rating elicitations under analysis, as described above. Every such elicitation is recorded in the specification grid and frozen corpus. The authors reviewed and verified all such output and take full responsibility for the content.

Declaration of competing interest

S.C. and R.C. each declare no competing interests.

Funding

This research received no external funding.

Ethics

This study did not involve human participants, human tissue, or animal subjects, and used no personally identifiable or sensitive personal data. Ethical approval was therefore not required. The analysis uses only synthetic firm profiles, publicly available corporate financial disclosures (SEC filings), and the authors' own model elicitations.

Data availability

All data and code required to reproduce the study are openly available in a deterministic, offline reproducibility capsule (MIT license) provided as anonymized supplementary material with this submission; the public repository and its citable DOI are withheld here for double-blind review and will be provided on acceptance. Within the capsule, `data/frozen/main/` contains the pre-registered 90-item firm battery (`battery_90.json`, 45 credit-health + 45 directional items), the specification grid, the prompt paraphrases, and the rating scale. `data/raw/main/` holds the frozen response corpus; `data/panel/panel.parquet` is the analysis panel, and `data/frozen/realarm/` with `data/raw/realarm/` contain the perturbed and fingerprint-screened real-firm arm (anonymized perturbed battery, sealed issuer crosswalk, rating provenance with verbatim 10-K quotation, permanent SEC URL, and document hash per issuer, and the arm/comparator/fingerprint model-output panels). The flagship model-tier arm (`gpt-5.4` + `gemini-3.1-pro-preview` on the frozen 12-specification grid) is in `capture/flagship_runs/`, fixed by `MANIFEST_FLAGSHIP.sha256`. The reasoning-architecture arm (`deepseek-r1` on the same grid) is in `capture/reasoning_runs/`, fixed by `MANIFEST_REASONING.sha256`. Analysis code is in `analysis/`, outputs and figures in `results/`, and SHA-256 manifests in `manifest/`. A single offline entry point, `reproduce.sh`, regenerates every reported number, table, and figure, with no model calls and no API keys. A self-contained companion, `REPRODUCIBILITY_DOCUMENTATION.pdf` (provided with this submission), documents the complete archive, covering data inputs and provenance, every analysis script and its outputs, the one-command reproduction, the verification and determinism procedure, and the pinned environment, so that the study can be understood and re-run from that single file. The underlying SEC filings (Form 10-K and company-facts XBRL) are publicly available through the SEC EDGAR system.

References

Altman, E.I., 1968. Financial ratios, discriminant analysis and the prediction of corporate bankruptcy. The Journal of Finance 23, 589–609. doi:10.1111/j.1540-6261.1968.tb00843.x.

Atil, B., Aykent, S., Chittams, A., Fu, L., Passonneau, R.J., Radcliffe, E., Rajagopal, G.R., Sloan, A., Tudrej, T., Ture, F., Wu, Z., Xu, L., Baldwin, B., 2024. Non-determinism of "deterministic" LLM settings. Technical Report. arXiv:2408.04667.

Basel Committee on Banking Supervision, 2013. Regulatory Consistency Assessment Programme (RCAP): Analysis of risk-weighted assets for credit risk in the banking book. Technical Report BCBS 256. Bank for International Settlements.

Basel Committee on Banking Supervision, 2019. CRE20: Standardised approach: individual exposures. Technical Report. Basel Framework, Bank for International Settlements. In force 1 January 2023.

Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency, 2011. Supervisory guidance on model risk management. Technical Report. SR Letter 11-7 / OCC Bulletin 2011-12.

Bommasani, R., et al., 2021. On the opportunities and risks of foundation models. arXiv preprint arXiv:2108.07258 .

Brown, T.B., et al., 2020. Language models are few-shot learners, in: Advances in Neural Information Processing Systems (NeurIPS).

Camuffo, A., Gambardella, A., Kazemi, S., Malachowski, J., Pandey, A., 2026. Variance-aware LLM annotation for strategy research: Sources, diagnostics, and a protocol for reliable measurement. Technical Report. arXiv:2601.02370.

Cantor, R., Packer, F., 1997. Differences of opinion and selection bias in the credit rating industry. Journal of Banking & Finance 21, 1395–1417. doi:10.1016/S0378-4266(97)00024-1.

Chon, S., Kim, J., Kim, J., 2025. Multifaceted variability in LLM-driven stock recommendations. Finance Research Letters Forthcoming.

Cornaggia, J., Cornaggia, K.J., Israelsen, R.D., 2018. Credit ratings and the cost of municipal financing. Review of Financial Studies 31, 2038–2079. doi:10.1093/rfs/hhx094.

Devlin, J., Chang, M.W., Lee, K., Toutanova, K., 2019. BERT: Pre-training of deep bidirectional transformers for language understanding, in: Proceedings of NAACL-HLT.

Drinkall, F., Pierrehumbert, J.B., Zohren, S., 2025. Forecasting credit ratings: A case study where traditional methods outperform generative LLMs, in: Proceedings of the Joint Workshop of FinNLP, FNP, and LLMFinLegal (COLING 2025), pp. 118–133.

Farquhar, S., Kossen, J., Kuhn, L., Gal, Y., 2024. Detecting hallucinations in large language models using semantic entropy. Nature 630, 625–630. doi:10.1038/s41586-024-07421-0.

García-Llorente, C., Olmeda, I., 2026. Large language models for banking supervision: Reliability evidence from European systemic banks. Finance Research Letters doi:10.1016/j.frl.2026.110440.

Gelman, A., Loken, E., 2014. The statistical crisis in science. American Scientist 102, 460–465. doi:10.1511/2014.111.460.

Hansen, A.L., Kazinnik, S., 2024. Can ChatGPT decipher Fedspeak? SSRN Electronic Journal doi:10.2139/ssrn.4399406.

Hens, T., Nordlie, T., 2025. How good are LLMs in risk profiling? Finance Research Letters 85, 108102. doi:10.1016/j.frl.2025.108102.

Kadavath, S., Conerly, T., Askell, A., Henighan, T., et al., 2022. Language models (mostly) know what they know. arXiv preprint arXiv:2207.05221 .

Kaplan, J., McCandlish, S., Henighan, T., Brown, T.B., Chess, B., Child, R., Gray, S., Radford, A., Wu, J., Amodei, D., 2020. Scaling laws for neural language models. arXiv preprint arXiv:2001.08361 .

Kliger, D., Sarig, O., 2000. The information value of bond ratings. Journal of Finance 55, 2879–2902. doi:10.1111/0022-1082.00311.

Kuhn, L., Gal, Y., Farquhar, S., 2023. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation, in: International Conference on Learning Representations (ICLR).

Lin, S., Hilton, J., Evans, O., 2022. Teaching models to express their uncertainty in words. Transactions on Machine Learning Research (TMLR) .

Lopez-Lira, A., Tang, Y., 2023. Can ChatGPT forecast stock price movements? return predictability and large language models. arXiv preprint arXiv:2304.07619 .

Mizrahi, M., Kaplan, G., Malkin, D., Dror, R., Shahaf, D., Stanovsky, G., 2024. State of what art? a call for multi-prompt LLM evaluation. Transactions of the Association for Computational Linguistics 12. doi:10.1162/tacl_a_00681.

Ohlson, J.A., 1980. Financial ratios and the probabilistic prediction of bankruptcy. Journal of Accounting Research 18, 109–131. doi:10.2307/2490395.

OpenAI, 2023. GPT-4 technical report. arXiv preprint arXiv:2303.08774 .

Ouyang, L., et al., 2022. Training language models to follow instructions with human feedback, in: Advances in Neural Information Processing Systems (NeurIPS).

Sclar, M., Choi, Y., Tsvetkov, Y., Suhr, A., 2024. Quantifying language models' sensitivity to spurious features in prompt design, in: International Conference on Learning Representations (ICLR).

Simmons, J.P., Nelson, L.D., Simonsohn, U., 2011. False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant. Psychological Science 22, 1359–1366. doi:10.1177/0956797611417632.

Simonsohn, U., Simmons, J.P., Nelson, L.D., 2020. Specification curve analysis. Nature Human Behaviour 4, 1208–1214. doi:10.1038/s41562-020-0912-z.

Steegen, S., Tuerlinckx, F., Gelman, A., Vanpaemel, W., 2016. Increasing transparency through a multiverse analysis. Perspectives on Psychological Science 11, 702–712. doi:10.1177/1745691616658637.

Tang, T.T., 2009. Information asymmetry and firms' credit market access: Evidence from Moody's credit rating format refinement. Journal of Financial Economics 93, 325–351. doi:10.1016/j.jfineco.2008.07.007.

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L., Polosukhin, I., 2017. Attention is all you need, in: Advances in Neural Information Processing Systems (NeurIPS).

Vickers, A.J., Elkin, E.B., 2006. Decision curve analysis: a novel method for evaluating prediction models. Medical Decision Making 26, 565–574. doi:10.1177/0272989X06295361.

Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., Zhou, D., 2023. Self-consistency improves chain of thought reasoning in language models, in: International Conference on Learning Representations (ICLR).

Wu, S., Irsoy, O., Lu, S., Dabravolski, V., Dredze, M., Gehrmann, S., Kambadur, P., Rosenberg, D., Mann, G., 2023. BloombergGPT: A large language model for finance. arXiv preprint arXiv:2303.17564 .

Xie, Q., Han, W., Zhang, X., Lai, Y., Peng, M., Lopez-Lira, A., Huang, J., 2023. PIXIU: A large language model, instruction data and evaluation benchmark for finance, in: Advances in Neural Information Processing Systems (NeurIPS).